

COHOMOLOGY OF DISCRETE GROUPS

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Original: Cohomologie des groupes discrets

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INTRODUCTION

Let Γ be a *discrete subgroup* of a locally compact group G . When G is a connected real Lie group, with a maximal compact subgroup K , the space $T = G/K$ is homeomorphic to a Euclidean space $\mathbf{R}^n$ and the action of Γ on T can be used for the study of the cohomology groups $H^q(\Gamma)$ of Γ . If for example Γ is torsion-free, it operates freely on T , and the $H^q(\Gamma)$ are identified with the corresponding cohomology groups of the manifold $X_\Gamma = \Gamma \backslash T$; the *cohomological dimension* of Γ is $\leq n$. If in addition G/Γ is compact, the Gauss-Bonnet formula, applied to X_Γ , allows one to express the Euler-Poincaré characteristic $\chi(\Gamma)$ of Γ as the volume $\mu_G(G/\Gamma)$ of G/Γ with respect to a certain invariant measure μ_G on G , independent of Γ ; we can call μ_G the *Euler-Poincaré measure* of G .

The above applies in particular to the *group of integral points* of an algebraic group L over the field $\mathbf{Q}$: we take for G the group $L(\mathbf{R})$ of real points of L . On the other hand, this does not apply to the more general groups which are the *S-arithmetic groups*, where one allows non-trivial denominators. Thus, the group $\Gamma = \mathrm{SL}_2(\mathbf{Z}[\frac{1}{p}])$ is not a discrete subgroup of $\mathrm{SL}_2(\mathbf{R})$; it is a discrete subgroup of the product

$$G = \mathrm{SL}_2(\mathbf{R}) \times \mathrm{SL}_2(\mathbf{Q}_p),$$

where $\mathbf{Q}_p$ is the field of p -adic numbers. The goal of this memoir is

to extend to such products G the well-known results recalled at the beginning. The essential tool, in the p -adic case, is the *Bruhat-Tits* building which renders the same services as the space G/K of the real theory: like it, it is contractible, and the group G operates on it properly; when $G = \mathrm{SL}_2(\mathbf{Q}_p)$, it is simply the *tree* defined in [38], Chap. II.

There are three sections. Here is their content:

Section 1 is preliminary; the group G does not appear in it. Nos. 1.1 to 1.4 recall various standard results and definitions on the cohomology of a discrete group Γ , its cohomological dimension (denoted $\mathrm{cd}(\Gamma)$), the finiteness property (FL) and the Euler-Poincaré characteristic $\chi(\Gamma)$; the connection with topology is indicated in No. 1.5. When Γ acts on a contractible CW-complex T , there exist certain simple relations between the cohomology of Γ and that of the stabilizers Γ_σ of the cells σ of T ; these relations, which were pointed out to me by Quillen, play an essential role in Sections 2 and 3; they are given in No. 1.6. No. 1.7 contains the proof, by topological means, of the formula $\mathrm{cd}(\Gamma) = \mathrm{cd}(\Gamma')$, valid when Γ is torsion-free and Γ' of finite index in Γ . No. 1.8 is devoted to “virtual” definitions: for example, a group Γ is said to be of virtual cohomological dimension $\leq n$ if it contains a subgroup of finite index Γ' such that $\mathrm{cd}(\Gamma') \leq n$. The usefulness of such notions comes from the fact that the most natural arithmetic groups, such as $\mathrm{SL}_n(\mathbf{Z})$, have torsion and are therefore of infinite cohomological dimension; but their virtual cohomological dimension is finite. Section 1 concludes with the case of Coxeter groups, and in particular the determination of their Euler-Poincaré characteristic.

Section 2 contains mainly upper bounds (and some calculations) of cohomological dimensions. These are discrete subgroups Γ of groups $G = \prod G_\alpha$, where the G_α are either real Lie groups with a finite number of connected components, or semisimple (or reductive) groups over local fields. When G_α is of the second type, the action

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of Γ on the building of G_α allows us to reduce the study of the cohomology of Γ to that of certain discrete subgroups of the product of the other G_β .

By induction, we are thus reduced to the real case, which is well known. All of this is the subject of nos. 2.1 to 2.3. No. 2.4 applies these results to S -arithmetic groups (over a number field or over a function field over a finite field); in particular, we obtain the fact that any torsion-free finitely generated subgroup of $\mathrm{GL}_n(\mathbf{Q})$ is of finite cohomological dimension.

Section 3 proves the existence, on a group G of the type above, of an *Euler-Poincaré measure* μ_G having the property indicated at the beginning: if Γ is a discrete subgroup with compact quotient of G , we have $\chi(\Gamma) = \mu_G(G/\Gamma)$.

The real case is treated, as we have seen, by means of the Gauss-Bonnet formula (no. 3.2). The p -adic case uses the building T : if μ is a Haar measure on G , we set

$$\chi(\mu) = \sum_{\sigma \in \Sigma} (-1)^{\dim(\sigma)} \frac{1}{\mu(G_\sigma)};$$

in this formula, Σ denotes a system of representatives of the cells of T modulo the action of G , and G_σ denotes the stabilizer of σ , which is an open compact subgroup of G . The product $\mu_G = \chi(\mu)\mu$ is independent of the choice of μ and is the desired Euler-Poincaré measure (no. 3.3). When G is semisimple, μ_G is > 0 (resp. < 0) if the relative rank of G is even (resp. odd), cf. no. 3.4. The application of these results to S -arithmetic groups with compact quotient is made in no. 3.6. The case of a non-compact quotient, which is much more delicate, has just been treated by Harder [19]; his results are recalled in nos. 3.6 and 3.7. Section 3 ends with an application of Euler-Poincaré characteristics to the estimation of the denominators of the values of zeta functions at negative integers.

The main results of this work have been summarized in a Note in the Comptes-Rendus [37].

§1. COHOMOLOGICAL DIMENSION AND FINITENESS PROPERTIES

1.1. Projective resolutions and free resolutions.

Let R be a ring and let E be an R -module (left, to fix ideas). A *projective resolution* (P_n) of E is an exact sequence of R -modules and R -homomorphisms

$$\cdots \rightarrow P_n \rightarrow P_{n-1} \rightarrow \cdots \rightarrow P_1 \rightarrow P_0 \rightarrow E \rightarrow 0$$

such that P_n is *projective* for all $n \geq 0$; this is equivalent to saying that $\mathbf{P}_\bullet = (P_n)$ is an acyclic, projective complex, equipped with an isomorphism $H_0(\mathbf{P}_\bullet) \rightarrow E$.

We say that (P_n) is of *length* $\leq m$ if $P_n = 0$ for $n > m$. For such a resolution to exist, it is necessary and sufficient that E be of projective dimension $\leq m$, or equivalently that $\text{Ext}_R^q(E, F) = 0$ for all $q > m$ and all R -modules F ([13], p. 110, prop. 2.1); one can then even choose *free* modules for P_n , provided that $m \geq 1$ (this follows from the fact that, if P is projective, there exists a free module L such that $P \oplus L \simeq L$, cf. Bourbaki, *Alg.* II, §2, exerc. 3).

A resolution (P_n) is said to be of *finite type* (resp. *free*) if all the modules P_n are of finite type (resp. free); (P_n) is said to be *finite* if it is both of finite type and of finite length.

The following result is well known:

Proposition (1). *Let $0 \rightarrow E_q \rightarrow \cdots \rightarrow E_1 \rightarrow E_0 \rightarrow E \rightarrow 0$ be an exact sequence of R -modules. For all i between 0 and q , let*

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$(P_{i,n})$ a projective resolution of E_i . There then exists a projective resolution (P_n) of E such that

$$P_n \simeq \coprod_{i+j=n} P_{i,j}$$

for all $n \geq 0$.

An argument by induction allows us to reduce to the case $q = 1$. We then have an exact sequence

$$0 \rightarrow E_1 \xrightarrow{f} E_0 \rightarrow E \rightarrow 0.$$

We “lift” the morphism f to a morphism of resolutions

$$(f_n): (P_{1,n}) \rightarrow (P_{0,n}),$$

and we take for (P_n) the “mapping cylinder” of (f_n) , cf. for example [13], p. 73, exercise 3. We indeed have $P_n = P_{1,n-1} \oplus P_{0,n}$ for all $n \geq 0$, and (P_n) is a projective resolution of E .

COROLLARY. *Let m be an integer ≥ 0 . If each E_i has a finite projective (resp. free) resolution of length $\leq m - i$, then E has a finite projective (resp. free) resolution of length $\leq m$.*

Suppose now that there exists a homomorphism $R \rightarrow k$ from the ring R to a field k . If L is a free R -module of finite type, isomorphic to R^d , we have $d = \dim_k(L \otimes_R k)$ which shows that d depends only on L ; the integer d is called the *rank* of the module L ; we will denote it by $\text{rg} L$.

If

$$0 \rightarrow L_n \rightarrow \cdots \rightarrow L_0 \rightarrow E \rightarrow 0$$

is a finite free resolution of a module E , we shall set

$$\chi_R(E) = \sum_{i=0}^n (-1)^i \text{rg}(L_i).$$

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We have

$$\chi_R(E) = \sum_{i=0}^n (-1)^i \dim_k \operatorname{Tor}_i^R(E, k),$$

which shows that the integer $\chi_R(E)$ does not depend on the chosen finite free resolution (L_i) ; we call it the *Euler-Poincaré characteristic* of E .

PROPOSITION 2. *Let $0 \rightarrow E_q \rightarrow \cdots \rightarrow E_1 \rightarrow E_0 \rightarrow E \rightarrow 0$ be an exact sequence of R -modules. Suppose that each E_i possesses a finite free resolution. Then the same holds for E , and we have*

$$\chi_R(E) = \sum_{i=0}^q (-1)^i \chi_R(E_i).$$

We apply Prop. 1 by taking for $(P_{i,n})$ a finite free resolution of E_i . This yields a finite free resolution (P_n) of E , with

$$P_n \simeq \coprod_{i+j=n} P_{i,j}$$

and

$$\chi_R(E) = \sum_{i,j} (-1)^{i+j} \operatorname{rg} P_{i,j} = \sum_i (-1)^i \chi_R(E_i).$$

REMARK. Let $K(R)$ be the *Grothendieck group* of the category of finitely generated projective R -modules; if P is such a module, let $[P]$ be its image in $K(R)$. The homomorphism $R \rightarrow k$ defines a homomorphism $e : K(R) \rightarrow K(k) = \mathbb{Z}$ such that $e([R]) = 1$.

If E is an R -module possessing a finite projective resolution (P_i) , let $[E]$ be the element of $K(R)$ defined by

$$[E] = \sum_i (-1)^i [P_i].$$

It is known that $[E]$ does not depend on the chosen resolution. In order for E

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admits a finite *free* resolution, it is necessary and sufficient that $[E]$ be a multiple of $[R]$, and we then have

$$[E] = \chi_R(E) \cdot [R], \quad e([E]) = \chi_R(E),$$

which gives a characterization of $\chi_R(E)$ by means of $[E]$. From this point of view, Prop. 2 merely expresses the additivity of the map $E \mapsto [E]$.

1.2. Cohomological dimension.

Let Γ be a group, and let $R = \mathbb{Z}[\Gamma]$ be its algebra over $\mathbb{Z}$. An R -module is also called a Γ -module. The ring R admits a homomorphism into a field (for example, the composition $R \rightarrow \mathbb{Z} \rightarrow \mathbb{Q}$); we can therefore apply to it the definitions and results of no. 1.1. In particular, if E is a Γ -module admitting a finite free resolution, the *Euler-Poincaré characteristic* $\chi_R(E)$ is defined; we also denote it by $\chi_\Gamma(E)$.

In all that follows, $\mathbb{Z}$ is endowed with the trivial Γ -module structure, i.e., such that $\gamma \cdot n = n$ for all $\gamma \in \Gamma$ and all $n \in \mathbb{Z}$.

PROPOSITION 3. *Let n be an integer ≥ 0 . The following conditions are equivalent:*

- (a) *The Γ -module $\mathbb{Z}$ has a projective resolution of length $\leq n$.*
- (b) *Every Γ -module which is free over $\mathbb{Z}$ has a projective resolution of length $\leq n$.*
- (c) *For every Γ -module M , and every integer $q > n$, the cohomology group $H^q(\Gamma, M)$ is reduced to 0.*

The equivalence of (a) and (c) follows from [13], p. 110, Prop. 2.1, taking into account that $H^q(\Gamma, M) = \operatorname{Ext}_R^q(\mathbb{Z}, M)$, with $R = \mathbb{Z}[\Gamma]$. The equivalence of (a) and (b) follows from the following lemma:

LEMMA 1. *Let (P_n) be a projective (resp. free) resolution of the Γ -module $\mathbb{Z}$, and let M be a Γ -module free over $\mathbb{Z}$. Then $(P_n \otimes M)$ is a projective (resp. free) resolution of M .*

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(Let us specify that, if A and B are two Γ -modules, we equip $A \otimes B$ with the "product" Γ -module structure, i.e., we have $s(a \otimes b) = s(a) \otimes s(b)$ if $s \in \Gamma$, $a \in A$, $b \in B$.)

Since M is free over $\mathbf{Z}$, the sequence

$$\cdots \rightarrow P_n \otimes M \rightarrow \cdots \rightarrow P_0 \otimes M \rightarrow M \rightarrow 0$$

is exact. Everything thus reduces to showing that $P \otimes M$ is a projective (resp. free) Γ -module if P is projective (resp. free); by decomposition into a direct sum, it suffices to consider the case where P is free of rank 1, with basis $\{p\}$; in this case, if M has a $\mathbf{Z}$ -basis (m_i) , one immediately verifies that $(p \otimes m_i)$ is a $\mathbf{Z}[\Gamma]$ -basis of $P \otimes M$.

DEFINITION. We call the cohomological dimension of the group Γ the least integer n such that the conditions of Prop. 3 are satisfied.

We denote by $\text{cd}(\Gamma)$ the cohomological dimension of Γ ; we have $0 \leq \text{cd}(\Gamma) \leq \infty$, with $\text{cd}(\Gamma) = \infty$ if and only if the conditions of Prop. 3 are not satisfied for any value of n .

We can also characterize $\text{cd}(\Gamma)$ as the supremum (finite or infinite) of the integers q such that there exists a Γ -module M with $H^q(\Gamma, M) \neq 0$.

DEFINITION. We say that Γ is of **type (FL)** if the Γ -module $\mathbf{Z}$ possesses a finite free resolution.

Let Γ be such a group. We have $\text{cd}(\Gamma) < \infty$. Furthermore, $\chi_\Gamma(\mathbf{Z})$ is defined; we call it the *Euler-Poincaré characteristic* of Γ , and we denote it by $\chi(\Gamma)$. By definition, we have

$$\chi(\Gamma) = \sum_i (-1)^i \text{rg}_\Gamma L_i,$$

if (L_i) is a finite free resolution of the Γ -module $\mathbf{Z}$.

PROPOSITION 4. Suppose Γ is of type (FL), and let M be a Γ -module.

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(a) If M is the inductive limit of Γ -modules (M_i) , and if q is an integer ≥ 0 , the canonical homomorphism from $\varinjlim H^q(\Gamma, M_i)$ to $H^q(\Gamma, M)$ is an isomorphism.

(b) If M is finitely generated over $\mathbf{Z}$, the same is true for $H^q(\Gamma, M)$ and we have

$$\sum_q (-1)^q \text{rg } H^q(\Gamma, M) = \chi(\Gamma) \cdot \text{rg } M.$$

(c) If M is $\mathbf{Z}$ -free of finite type, M has a finite $\mathbf{Z}[\Gamma]$ -free resolution, and we have

$$\chi_\Gamma(M) = \chi(\Gamma) \cdot \text{rg } M.$$

Let $L_\bullet = (L_n)$ be a finite $\mathbf{Z}[\Gamma]$ -free resolution of $\mathbf{Z}$. The groups $H^q(\Gamma, M)$ are the cohomology groups of the complex $\text{Hom}(L_\bullet, M)$, from which (a) and (b) follow immediately. Similarly, (c) follows from the fact that $L_\bullet \otimes M$ is a finite free resolution of M , cf. lemma 1.

REMARKS

1) Let k be a field, and let M be a $k[\Gamma]$ -module of finite dimension over k . The same argument as above shows that the $H^q(\Gamma, M)$ are finite-dimensional k -vector spaces, and that we have

$$\sum_q (-1)^q \dim_k H^q(\Gamma, M) = \chi(\Gamma) \cdot \dim_k M.$$

Furthermore, M has a finite $k[\Gamma]$ -free resolution, of Euler-Poincaré characteristic equal to $\chi(\Gamma) \cdot \dim_k M$.

2) One can show that the Γ -module $\mathbf{Z}$ has a finite free resolution of length equal to $\text{cd}(\Gamma)$.

QUESTION

Let Γ be a group such that the Γ -module $\mathbb{Z}$ has a finite projective resolution. Is the group Γ of type (FL)?

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1.3. Subgroups, extensions, amalgams.

Subgroups

PROPOSITION 5. *Let Γ' be a subgroup of the group Γ .*

- (a) *We have $\text{cd}(\Gamma') \leq \text{cd}(\Gamma)$.*
- (b) *If $\text{cd}(\Gamma) < \infty$ and if Γ' is of finite index in Γ , we have $\text{cd}(\Gamma') = \text{cd}(\Gamma)$.*
- (c) *If Γ is of type (FL) and if Γ' is of finite index in Γ , then Γ' is of type (FL) and we have $\chi(\Gamma') = (\Gamma : \Gamma')\chi(\Gamma)$.*

Since $\mathbf{Z}[\Gamma]$ is a free $\mathbf{Z}[\Gamma']$ -module of rank $(\Gamma : \Gamma')$, any free $\mathbf{Z}[\Gamma]$ -resolution (L_n) of $\mathbf{Z}$ is also a free $\mathbf{Z}[\Gamma']$ -resolution; hence (a). If, moreover, (L_n) is $\mathbf{Z}[\Gamma]$ -finite, and if $(\Gamma : \Gamma') < \infty$, we see that (L_n) is $\mathbf{Z}[\Gamma']$ -finite, and we have

$$\begin{aligned}\chi(\Gamma') &= \sum_n (-1)^n \text{rg}_{\Gamma'} L_n = \sum_n (-1)^n (\Gamma : \Gamma') \text{rg}_{\Gamma} L_n \\ &= (\Gamma : \Gamma') \chi(\Gamma),\end{aligned}$$

hence (c).

It remains to prove (b). Suppose then that $(\Gamma : \Gamma')$ is finite, and set $n = \text{cd}(\Gamma)$. By hypothesis, there exists a Γ -module M such that $H^n(\Gamma, M) \neq 0$. We will show that $H^n(\Gamma', M) \neq 0$; this will show that $\text{cd}(\Gamma') \geq n$, hence $\text{cd}(\Gamma') = n$ by (a).

The non-vanishing of $H^n(\Gamma', M)$ follows from the following lemma:

LEMMA 2 (Tate). *The corestriction homomorphism*

$$\text{Cor} : H^n(\Gamma', M) \rightarrow H^n(\Gamma, M)$$

is surjective if $n = \text{cd}(\Gamma)$.

The proof is the same as in the profinite case ([35], p. I-20, lemma 4). We identify $H^n(\Gamma', M)$ with $H^n(\Gamma, M^*)$, where M^* denotes the module

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...induced from M , the set of functions $f : \Gamma \rightarrow M$ such that $f(sx) = sf(x)$ if $s \in \Gamma'$, $x \in \Gamma$; we let Γ act on M^* by $(tf)(x) = f(xt)$. We define a surjective homomorphism $\pi : M^* \rightarrow M$ by

$$\pi(f) = \sum_{x \in \Gamma/\Gamma'} x \cdot f(x^{-1}),$$

cf. [35], p. I-13. By passing to cohomology, π gives the corestriction

$$\text{Cor} : H^q(\Gamma', M) = H^q(\Gamma, M^*) \rightarrow H^q(\Gamma, M).$$

The fact that Cor is surjective in dimension n then follows simply from the fact that the functor $E \mapsto H^n(\Gamma, E)$ is right exact, given the vanishing of $H^{n+1}(\Gamma)$.

REMARK. Suppose that Γ' is of finite index in Γ . According to (a) and (b), we have the following two possibilities:

- (b₁) $\text{cd}(\Gamma') = \text{cd}(\Gamma)$;
- (b₂) $\text{cd}(\Gamma') < \infty$ and $\text{cd}(\Gamma) = \infty$.

We shall see later (no. 1.7, Th. 1) that (b₂) can only occur if Γ has torsion.

Extensions

Proposition 6. *Let Γ' be a normal subgroup of Γ .*

- (a) *We have $\text{cd}(\Gamma) \leq \text{cd}(\Gamma') + \text{cd}(\Gamma/\Gamma')$.*
- (b) *If Γ' and Γ/Γ' are of type (FL), the same holds for Γ , and we have*

$$\chi(\Gamma) = \chi(\Gamma') \cdot \chi(\Gamma/\Gamma').$$

Suppose Γ' is of type (FL), and let

$$0 \rightarrow L'_n \rightarrow \cdots \rightarrow L'_0 \rightarrow \mathbf{Z} \rightarrow 0$$

be a finite $\mathbf{Z}[\Gamma']$ -free resolution of $\mathbf{Z}$. By tensor product with $\mathbf{Z}[\Gamma]$, we deduce a finite $\mathbf{Z}[\Gamma]$ -free resolution of the Γ -module

$$\mathbf{Z}[\Gamma] \otimes_{\mathbf{Z}[\Gamma']} \mathbf{Z} = \mathbf{Z}[\Gamma/\Gamma'] :$$

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$$0 \rightarrow L_n \rightarrow \cdots \rightarrow L_0 \rightarrow Z[\Gamma/\Gamma'] \rightarrow 0,$$

where $L_i = Z[\Gamma] \otimes_{\mathbf{Z}[\Gamma']} L'_i$. We therefore have $\chi_\Gamma(Z[\Gamma/\Gamma']) = \chi(\Gamma')$.

Suppose now Γ/Γ' is of type (FL), and let

$$0 \rightarrow L''_m \rightarrow \cdots \rightarrow L''_0 \rightarrow Z \rightarrow 0$$

be a finite free $Z[\Gamma/\Gamma']$ -resolution of Z . In view of the above, each L''_i admits a finite free $Z[\Gamma]$ -resolution; moreover, we have

$$\chi_\Gamma(L''_i) = \chi(\Gamma') \text{rg}_{\Gamma/\Gamma'}(L''_i).$$

Applying Propositions 1 and 2, we deduce that Z admits a finite free $Z[\Gamma]$ -resolution, and that we have

$$\chi(\Gamma) = \chi_\Gamma(Z) = \sum_i (-1)^i \chi_\Gamma(L''_i) = \chi(\Gamma') \cdot \chi(\Gamma/\Gamma'),$$

which proves (b).

Assertion (a) is proved in an analogous manner (or is deduced from the spectral sequence of group extensions, cf. [13], p. 350).

Amalgams

Let Γ_1 and Γ_2 be two groups containing as a subgroup the same group A , and let $\Gamma = \Gamma_1 *_A \Gamma_2$ be the amalgamated sum of Γ_1 and Γ_2 along A (cf. Bourbaki, *Alg.* I, new ed., p. 83 or [38], I, §1). Let M be a Γ -module. By a result of Lyndon [28] (see also Swan [44], Thm. 2.3), we have a “Mayer-Vietoris” exact sequence:

$$\cdots \rightarrow H^q(\Gamma, M) \xrightarrow{\alpha} H^q(\Gamma_1, M) \oplus H^q(\Gamma_2, M) \xrightarrow{\beta} H^q(A, M) \rightarrow H^{q+1}(\Gamma, M) \rightarrow \cdots$$

where $\alpha = (\text{Res}, \text{Res})$ and $\beta = (\text{Res}, -\text{Res})$. From this we deduce:

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PROPOSITION 7. *Let n be an integer. If $\text{cd}(\Gamma_1) \leq n$, $\text{cd}(\Gamma_2) \leq n$ and $\text{cd}(A) \leq n - 1$, we have $\text{cd}(\Gamma_1 *_A \Gamma_2) \leq n$.*

Furthermore:

PROPOSITION 8. *If A , Γ_1 and Γ_2 are of type (FL), the same holds for $\Gamma_1 *_A \Gamma_2$, and we have*

$$\chi(\Gamma_1 *_A \Gamma_2) = \chi(\Gamma_1) + \chi(\Gamma_2) - \chi(A).$$

If B is a subgroup of $\Gamma = \Gamma_1 *_A \Gamma_2$, let us denote by $\mathbb{Z}[\Gamma/B]$ the free abelian group with base the set Γ/B ; we equip $\mathbb{Z}[\Gamma/B]$ with the unique Γ -module structure extending the natural action of Γ on Γ/B . According to Lyndon and Swan ([44], Lemma 2.1), we have an exact sequence

$$0 \rightarrow \mathbb{Z}[\Gamma/A] \rightarrow \mathbb{Z}[\Gamma/\Gamma_1] \oplus \mathbb{Z}[\Gamma/\Gamma_2] \rightarrow \mathbb{Z} \rightarrow 0.$$

(The exactness of this sequence simply expresses the fact that the tree associated with the amalgam Γ ([38], I-50) is acyclic.)

By hypothesis, the A -module $\mathbb{Z}$ admits a finite free resolution (L_n) . By tensor product with $\mathbb{Z}[\Gamma]$, we deduce a finite free resolution of the Γ -module $\mathbb{Z}[\Gamma/A]$ and we see at the same time that the Euler-Poincaré characteristic of this module is equal to $\chi(A)$. We have analogous results for $\mathbb{Z}[\Gamma/\Gamma_1]$ and $\mathbb{Z}[\Gamma/\Gamma_2]$. We then conclude by applying Prop. 1 and 2.

1.4. Examples.

a) We have $\text{cd}(\Gamma) = 0$ if and only if $\Gamma = \{1\}$. Indeed, if $\text{cd}(\Gamma) = 0$, $\mathbb{Z}$ is Γ -projective, and the canonical homomorphism $\mathbb{Z}[\Gamma] \xrightarrow{\alpha} \mathbb{Z}$ admits a section. There exists therefore an element $u \in \mathbb{Z}[\Gamma]$, invariant under Γ , and such that $\alpha(u) = 1$; it is immediate that this implies $\Gamma = \{1\}$.

More generally, let A be a commutative ring $\neq 0$. Let us define $\text{cd}_A(\Gamma)$ as the projective dimension of the $A[\Gamma]$ -module A ; it is also the

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upper bound of the integers q such that there exists an $A[\Gamma]$ -module M with $H^q(\Gamma, M) \neq 0$. The same argument as above shows that $\text{cd}_A(\Gamma) = 0$ if and only if Γ is a finite group of order invertible in A .

b) A finite group of order $\neq 1$ has infinite cohomological dimension. This is a well-known result, cf. for example [13], p. 265, exerc. 14.

By Prop. 5, we deduce from this that *every group of finite cohomological dimension is torsion-free*.

c) We have $\text{cd}(\Gamma) \leq 1$ if and only if Γ is a free group; this result, conjectured for a long time, was recently proved by Stallings [43] and Swan [44]; the latter even proved that, if Γ is torsion-free, and if $\text{cd}_A(\Gamma) \leq 1$ for some ring $A \neq 0$, then Γ is free.

If Γ is free of finite rank n (i.e., admits a basis of n elements), Γ is of type (FL) and we have $\chi(\Gamma) = 1 - n$.

d) Groups of cohomological dimension ≤ 2 . Let us write Γ in the form F/R , where F is free with basis $(x_i)_{i \in I}$ and where R is a normal subgroup of F . If (R, R) denotes the derived subgroup of R , the quotient $R/(R, R)$ is a Γ -module. Furthermore, we have an exact sequence of Γ -modules (cf. Swan [44], th. 1.4):

$$0 \rightarrow R/(R, R) \xrightarrow{\theta} X \xrightarrow{\partial} \mathbb{Z}[\Gamma] \rightarrow \mathbb{Z} \rightarrow 0$$

where X is $\mathbb{Z}[\Gamma]$ -free with basis $(e_i)_{i \in I}$. We have $\partial(e_i) = x_i - 1$, and θ is defined by means of the “partial derivatives” $\partial/\partial x_i$, cf. Swan, *loc. cit.* From this we deduce that $\text{cd}(\Gamma) \leq 2$ if and only if $R/(R, R)$ is $\mathbb{Z}[\Gamma]$ -projective.

Suppose in particular that I is finite, and that R is generated as a normal subgroup by an element r which is not an m -th power for any $m \geq 2$. Under these hypotheses, Lyndon [28] showed that the image of r in $R/(R, R)$ is a basis of this $\mathbb{Z}[\Gamma]$ -module. We then have $\text{cd}(\Gamma) \leq 2$, the group Γ is of type (FL), and $\chi(\Gamma) = 2 - \text{Card}(I)$.

e) A finitely generated torsion-free nilpotent group is of type (FL). This is proved by induction on the class of the group, using the

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prop. 6. (If Γ is such a group, one can also embed Γ into a simply connected nilpotent real Lie group G such that G/Γ is compact, and use prop. 18 of no. 2.1; this shows at the same time that $\text{cd}(\Gamma)$ is equal to $\dim(G)$.)

f) *Every torsion-free “arithmetic” subgroup of a semi-simple algebraic group is of type (FL)* (Raghunathan [33]); we will see later (no. 2.4, th. 4) that this result extends to “S-arithmetic” groups.

g) *Every subgroup of $\text{GL}_n(\mathbf{Q})$ which is finitely generated and torsion-free is of finite cohomological dimension*, cf. no. 2.4, th. 5.

1.5. Cohomological dimension and $K(\Gamma, 1)$ spaces.

Let Γ be a group. We know that there exists a CW-complex X (in the sense of J.H.C. Whitehead [51]) which is a “ $K(\Gamma, 1)$ space”, i.e., which satisfies the following conditions:

- (a) X is connected and non-empty;
- (b) the fundamental group $\pi_1(X)$ of X relative to a basepoint x is isomorphic to Γ ;
- (c) the higher homotopy groups $\pi_n(X)$ are zero for $n \geq 2$.

The conditions (a) and (c) are equivalent to saying that the universal covering $\bar{X}$ of X (relative to x) is *contractible*.

If X satisfies these properties, the complex $C_\bullet(\bar{X})$ of cellular chains of $\bar{X}$ is a *free resolution* of the Γ -module $\mathbf{Z}$; more precisely, let Σ be the set of cells of X , and for every $\sigma \in \Sigma$, let $\bar{\sigma}$ be a lift of σ to $\bar{X}$; if we equip each $\bar{\sigma}$ with an orientation, we obtain an element $\bar{\sigma}$ of $C_\bullet(\bar{X})$ and the family $(\bar{\sigma})$ is a *basis* of the $\mathbf{Z}[\Gamma]$ -module $C_\bullet(\bar{X})$.

It follows that, if M is a Γ -module, the $H^q(\Gamma, M)$ are isomorphic to $H^q(\text{Hom}_\Gamma(C_\bullet(\bar{X}), M))$, in other words to the cohomology groups $H^q(X, \tilde{M})$ of the space X with coefficients in the *local system* $\tilde{M}$ defined by M . We immediately deduce from this:

PROPOSITION 9. (a) *We have $\text{cd}(\Gamma) \leq \dim(X)$.*

(b) *If X is a finite complex, Γ is of type (FL) and $\chi(\Gamma)$ is equal to the Euler-Poincaré characteristic $\chi(X)$ of X .*

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(Note that, in case (b), the group Γ is *finitely presented*.)

La Proposition 9 provides a convenient way to prove that a given group Γ is of cohomological dimension $\leq n$: it suffices to make Γ act freely and properly on a contractile space $\bar{X}$ chosen such that the quotient $X = \bar{X}/\Gamma$ is a CW-complex of dimension $\leq n$. We will see many examples of this in the following.

For $n \geq 3$, we have a converse (cf. Eilenberg-Ganea [16]):

PROPOSITION 10. *Let n be an integer ≥ 3 , and let Γ be a group such that $\text{cd}(\Gamma) \leq n$.*

- (a) *There exists a CW-complex X of dimension $\leq n$ which is a $K(\Gamma, 1)$ space.*
- (b) *If Γ is finitely presented and of type (FL), one can choose X to be a finite complex.*

Let Y be a CW-complex which is a $K(\Gamma, 1)$ space. Since $\text{cd}(\Gamma) \leq n$, we have $H^q(Y, \tilde{M}) = 0$ for all $q > n$ and all local systems $\tilde{M}$ on Y . By a result of Wall ([48], Th. E), it follows that Y has the same homotopy type as a CW-complex X of dimension $\leq n$, which establishes (a).

Assertion (b) is proved in a similar way. One can, for example, apply Theorem F of [48], after having shown that Γ satisfies “condition F_n ”; one can also reduce to Th. 4 of [49]. Since we will not use (b) in the sequel, we leave the details of these proofs to the reader.

REMARK

One may wonder if Prop. 10 remains valid for $n < 3$. This is obviously the case if $n = 0$; it is also the case if $n = 1$ thanks to the Stallings-Swan theorem, cf. $n^\circ 4$, c). For $n = 2$, the question (in the form (a), to fix ideas) is reformulated as follows:

If $\text{cd}(\Gamma) \leq 2$, can one write Γ in the form F/R , with F free, in such a way that there exists a family (r_i) of elements of R , generating R as a normal subgroup of F , and such that the images of the r_i in

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$\dots R/(R, R)$ form a *basis* of $R/(R, R)$ as a $Z[\Gamma]$ -module? This seems unlikely, but I do not know of any counterexample.

1.6 Cellular actions.

Let T be a CW-complex on which the group Γ acts. We make the following two assumptions:

- (i) T is *acyclic*, i.e., $H_0(T, Z) = Z$ and $H_q(T, Z) = 0$ for all $q \geq 1$ (this is in particular the case if T is contractible).
- (ii) Γ acts *cellularly* on T ; this means that the image of a cell of T under an element of Γ is a cell of T .

If σ is a cell of T , let us denote by Γ_σ the set of $\gamma \in \Gamma$ such that $\gamma\sigma = \sigma$. Let, on the other hand, Σ be a set of representatives of the cells of T modulo the action of Γ . The following proposition allows, to some extent, to pass from the Γ_σ ($\sigma \in \Sigma$) to Γ itself:

PROPOSITION 11 (Quillen). (a) *We have $\text{cd}(\Gamma) \leq \sup_{\sigma \in \Sigma} (\dim(\sigma) + \text{cd}(\Gamma_\sigma))$.*

(b) *If Σ is finite and if the Γ_σ are of type (FL), then Γ is of type (FL), and we have*

$$\chi(\Gamma) = \sum_{\sigma \in \Sigma} (-1)^{\dim(\sigma)} \chi(\Gamma_\sigma).$$

Let $n = \dim(T)$. We can assume n is finite, for otherwise there is nothing to prove. Since T is acyclic, we have an exact sequence of Γ -modules

$$0 \rightarrow C_n(T) \rightarrow \dots \rightarrow C_0(T) \rightarrow Z \rightarrow 0,$$

where $C_q(T)$ is the group of *cellular chains* of degree q of T ; if T_q denotes the q -skeleton of T , we have

$$C_q(T) = H_q(T_q \text{ mod. } T_{q-1}; Z).$$

The structure of the Γ -module $C_q(T)$ is easy to determine:

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Let Σ_q be the subset of Σ formed by the cells of dimension q ; if $\sigma \in \Sigma_q$, and if $\gamma \in \Gamma_\sigma$, we set $\varepsilon_\sigma(\gamma) = 1$ (resp. -1) if γ preserves (resp. reverses) the orientation of σ . We thus obtain a homomorphism $\varepsilon_\sigma : \Gamma_\sigma \rightarrow \{\pm 1\}$, hence an action of Γ_σ on $\mathbf{Z}$; let Z_σ be the Γ_σ -module thus defined, and let $Z_\sigma^* = Z[\Gamma] \otimes_{Z[\Gamma_\sigma]} Z_\sigma$ be the corresponding induced Γ -module.

Lemma (3). (i) *The Γ -module Z_σ^* is of projective dimension $\leq \text{cd}(\Gamma_\sigma)$; if Γ_σ is of type (FL), Z_σ^* admits a finite free resolution, and its Euler-Poincaré characteristic is equal to $\chi(\Gamma_\sigma)$.*

(ii) *$C_q(T)$ is isomorphic to the direct sum of the Z_σ^* , $\sigma \in \Sigma_q$.*

Prop. 3 shows that the Γ_σ -module Z_σ has a projective resolution of length $\leq \text{cd}(\Gamma_\sigma)$; by tensor product with $Z[\Gamma]$, we deduce a projective resolution of the Γ -module Z_σ^* of length $\leq \text{cd}(\Gamma_\sigma)$. Similarly, if Γ_σ is of type (FL), one can find a finite free resolution of Z_σ , and its Euler-Poincaré characteristic is equal to $\chi(\Gamma_\sigma)$, cf. prop. 4; by tensor product with $Z[\Gamma]$, we deduce the same result for the Γ -module Z_σ^* . Hence (i). Assertion (ii) follows simply from the fact that every cell of T can be transformed by Γ into one and only one cell of Σ .

Let us now return to prop. 11. Let c_q be the projective dimension of the Γ -module $C_q(T)$. By Lemma 3, we have

$$c_q \leq \sup_{\sigma \in \Sigma_q} \text{cd}(\Gamma_\sigma) \quad \text{for all } q \geq 0.$$

Applying the cor. to prop. 1, we deduce

$$\text{cd}(\Gamma) \leq \sup_{q \leq n} \left(q + \sup_{\sigma \in \Sigma_q} \text{cd}(\Gamma_\sigma) \right) = \sup_{\sigma \in \Sigma} (\dim(\sigma) + \text{cd}(\Gamma_\sigma)),$$

which proves (a).

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In case (b), Lemma 3 shows that the $C_q(T)$ have finite free resolutions, whence the same result for the Γ -module $\mathbf{Z}$ by the corollary to Proposition 1. Moreover, Proposition 2 shows that one has

$$\chi(\Gamma) = \chi_\Gamma(\mathbf{Z}) = \sum_q (-1)^q \chi_\Gamma(C_q(T)),$$

whence, by applying Lemma 3,

$$\chi(\Gamma) = \sum_{\sigma \in \Sigma} (-1)^{\dim(\sigma)} \chi_\Gamma(\mathbf{Z}_\sigma^*) = \sum_{\sigma \in \Sigma} (-1)^{\dim(\sigma)} \chi(\Gamma_\sigma),$$

which completes the proof of (b).

REMARKS

- 1) One easily deduces from the exact sequence

$$0 \rightarrow C_n(T) \rightarrow \cdots \rightarrow C_0(T) \rightarrow \mathbf{Z} \rightarrow 0$$

and from Lemma 3, the existence, for every Γ -module M , of a *spectral sequence* having as E_1 term the (graded) product of the $H^\bullet(\Gamma_\sigma, \mathbf{Z}_\sigma \otimes M)$, $\sigma \in \Sigma$, and abutting to the graded group associated with $H^\bullet(\Gamma, M)$. This gives another proof of (a).

- 2) When T is *contractible* and the Γ_σ are *reduced to* $\{1\}$, the space $X = T/\Gamma$ can be endowed with a quotient CW-complex structure of that of T , and one shows that T identifies with the universal covering of X ; we recover the situation of the preceding section.

EXAMPLE

Let us apply Proposition 11 to the *tree* T associated with an *amalgam* $\Gamma = \Gamma_1 *_A \Gamma_2$, cf. [38], chap. I, no. 4.1. The fundamental domain of Γ is a segment $P_1 P_2$; we can therefore take for Σ the vertices P_1, P_2 and the edge $P_1 P_2$; the corresponding stabilizers Γ_σ are Γ_1, Γ_2 , and A . Proposition 11 then gives

$$\text{cd}(\Gamma) \leq \text{Sup}(\text{cd}(\Gamma_1), \text{cd}(\Gamma_2), 1 + \text{cd}(A))$$

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and

$$\chi(\Gamma) = \chi(\Gamma_1) + \chi(\Gamma_2) - \chi(A)$$

if A, Γ_1 and Γ_2 are of type (FL).

One thus recovers Prop. 7 and 8. Moreover, if Γ' is a subgroup of Γ , Prop. 11, applied to the action of Γ' on T , shows that we have $\text{cd}(\Gamma') \leq n$ if the intersections of Γ' with the conjugates of Γ_1 and Γ_2

(resp. of A) are of cohomological dimension $\leq n$ (resp. $\leq n - 1$). The same method applies to more complicated amalgams, cf. [38], Chap. I, §5.

1.7. Subgroups of finite index.

THEOREM 1. *Let Γ' be a subgroup of finite index of a group Γ . If Γ is torsion-free, we have $\text{cd}(\Gamma') = \text{cd}(\Gamma)$.*

By Prop. 5, the equality $\text{cd}(\Gamma') = \text{cd}(\Gamma)$ is true if $\text{cd}(\Gamma) < \infty$. Everything therefore reduces to showing that, if $\text{cd}(\Gamma')$ is finite, the same holds for $\text{cd}(\Gamma)$. This will follow from the following proposition:

PROPOSITION 12. *Suppose that $\text{cd}(\Gamma')$ and $(\Gamma : \Gamma')$ are finite. We can then make Γ operate cellularly on an acyclic CW-complex Z of finite dimension such that, for every cell σ of Z , the stabilizer Γ_σ of σ in Γ is finite.*

(If Γ is torsion-free, the Γ_σ are reduced to $\{1\}$, and Prop. 11 shows that $\text{cd}(\Gamma) \leq \dim(Z) < \infty$.)

Let $n = \sup(3, \text{cd}(\Gamma'))$. By Prop. 10, there exists a CW-complex X of dimension n which is a $K(\Gamma', 1)$ space. Let T be the universal cover of such a space X , relative to a basepoint. Let us equip T with the CW-complex structure which is the preimage of that of X ([51], §5, (N)). The group Γ' operates freely and cellularly on T ; we have $T/\Gamma' = X$.

Let

$$Y = \Gamma \times^{\Gamma'} T$$

be the Γ -space induced from the Γ' -space T ; by definition, Y is the quotient of $\Gamma \times T$ (Γ having the discrete topology) by Γ' operating by

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$$a \cdot (\gamma, t) = (\gamma a^{-1}, at) \quad \text{if } a \in \Gamma', \gamma \in \Gamma, t \in T;$$

furthermore, the transform under $\delta \in \Gamma$ of the class of (γ, t) is the class of $(\delta\gamma, t)$. (Another definition of Y : it is the covering of X associated with the Γ' -set Γ .)

The space T is identified with a subspace of Y via $t \mapsto (1, t)$; the action of Γ on Y extends that of Γ' on T . If $\gamma_1, \gamma_2 \in \Gamma$, we have $\gamma_1 T = \gamma_2 T$ if and only if $\gamma_1 \Gamma' = \gamma_2 \Gamma'$, which gives meaning to the expression aT , for $a \in \Gamma/\Gamma'$. The space Y is the disjoint union of the aT ($a \in \Gamma/\Gamma'$); in particular, the aT are the connected components of Y . The CW-complex structure of T transfers to the aT , and defines a CW-complex structure on Y , invariant under Γ .

Let now

$$Z = \prod_{a \in \Gamma/\Gamma'} aT$$

be the *product* CW-complex of the CW-complexes aT (note that the topology of Z is not necessarily the product topology: it is the inductive limit of the topologies of its finite subcomplexes, cf. [51]). Since Γ operates cellularly on Y , and permutes the aT among themselves, it operates cellularly on Z . Furthermore:

(i) Z is contractible. Indeed, each of the complexes aT is isomorphic to T , hence contractible.

(ii) For every cell σ of Z , the stabilizer Γ_σ of σ is finite. Indeed, we have $\sigma = \prod \sigma_\alpha$, where σ_α is a cell of aT . If γ belongs to Γ_σ , γ permutes the σ_α among themselves. If, moreover, γ belongs to Γ' , we have $\gamma\sigma_1 = \sigma_1$, since γ maps T into itself. Since Γ' operates freely on the cells of T , this implies $\gamma = 1$; we therefore have $\Gamma_\sigma \cap \Gamma' = \{1\}$, and since Γ' is of finite index in Γ , this indeed shows that Γ_σ is finite.

The CW-complex Z therefore answers the question.

REMARK

The preceding proof can be transcribed in purely algebraic terms:

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Suppose $\text{cd}(\Gamma')$ is finite. Let L be a $Z[\Gamma']$ -projective resolution of Z of finite length, and let $M = Z[\Gamma] \otimes_{Z[\Gamma']} L$ be the corresponding “induced” complex. This complex contains L as a subcomplex, and one has

$$M = \prod_{\alpha \in \Gamma/\Gamma'} \alpha L.$$

The subcomplexes αL are permuted among themselves by the action of Γ . Let us form their tensor product (relative to a chosen total order¹ on Γ/Γ'):

$$N = \bigotimes_{\alpha \in \Gamma/\Gamma'} (\alpha L).$$

The group Γ operates in a natural way on N (taking into account the “signs” due to the fact that Γ does not preserve the chosen order structure on Γ/Γ'). It is clear that N is acyclic (Künneth formula) and of finite length; if Γ is torsion-free, an argument analogous to the one used above shows that N is $Z[\Gamma]$ -projective. Theorem 1 follows from this (see [44], §9, for more details).

This “algebrisation” has two advantages:

- i) One can replace Z there by any commutative ring A . One thus obtains the following result (cf. [44], th. 9.2): *Let Γ' be a subgroup of finite index of Γ ; suppose that Γ has no A -torsion, i.e., that if $n \geq 1$ is the order of an element of Γ , n is invertible in A ; then $\text{cd}_A(\Gamma') = \text{cd}_A(\Gamma)$.*
- ii) One can employ the same method for the cohomology of profinite groups (cf. [35]), on condition of replacing the $Z[\Gamma]$ -projective modules by “pro-free” modules relatively to the completed algebra of the profinite group Γ under consideration. One thus sees that, if Γ has no

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no element of order p (p prime) and if Γ' is an open subgroup of Γ , we have $\text{cd}_p(\Gamma') = \text{cd}_p(\Gamma)$, a result which had been proved in [36] by means of Steenrod powers.

1.8. Virtual notions.

We shall say that a group Γ is *virtually torsion-free* if there exists a subgroup of Γ which is *torsion-free* and of finite index in Γ .

EXAMPLES

A classical argument, due to Minkowski [31], shows that every S -arithmetic group is virtually torsion-free. The same is true, more generally, for any finitely generated subgroup of a linear group over a field of characteristic zero (cf. for example [7], 17.7).

DEFINITION. Let Γ be a virtually torsion-free group. We call the *virtual cohomological dimension* of Γ , denoted by $\text{vcd}(\Gamma)$, the cohomological dimension of the torsion-free subgroups of finite index in Γ .

(This definition is well-defined because all these subgroups have the same cohomological dimension, by virtue of Th. 1.)

REMARKS

- a) We have $\text{vcd}(\Gamma) = 0$ if and only if Γ is *finite*.
- b) If Γ' is a subgroup of Γ , we have $\text{vcd}(\Gamma') \leq \text{vcd}(\Gamma)$ and there is equality if $(\Gamma : \Gamma')$ is finite.
- c) If Γ is torsion-free, we have $\text{vcd}(\Gamma) = \text{cd}(\Gamma)$. On the other hand, if Γ has torsion, we have $\text{cd}(\Gamma) = \infty$, while $\text{vcd}(\Gamma)$ can be finite.

DEFINITION. We say that Γ is *of type (VFL)* (or that Γ is *virtually of type (FL)*) s'il existe un sous-groupe d'indice fini de Γ qui est *of type (FL)*.

Let Γ be such a group. It is clear that Γ is virtually torsion-free, and that $\text{vcd}(\Gamma)$ is *finite*. Let us choose a subgroup Γ' of Γ which is of type (FL), and set (cf. Wall [47]):

$$\chi(\Gamma) = \frac{1}{(\Gamma : \Gamma')} \chi(\Gamma').$$

¹One can avoid such a choice; one must then define a notion of “tensor product of complexes” which does not depend on an order relation on the set of indices; we leave the details to the reader.

According to Prop. 5 (c), the rational number $\chi(\Gamma)$ thus defined does not depend on the choice of Γ' . It is called the *Euler-Poincaré characteristic* of Γ .

EXAMPLES

- 1) A *finite* group Γ is of type (VFL) and $\chi(\Gamma) = 1/\text{Card}(\Gamma)$.
- 2) Let $\Gamma = \text{SL}_2(\mathbf{Z})$. It is known that Γ is the sum of the cyclic group of order 4 generated by $x = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and of the cyclic group of order 6 generated by $y = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$, these two groups being amalgamated along their intersection $\{\pm 1\}$. One easily deduces from this that the *derived group* $\Gamma' = (\Gamma, \Gamma)$ of Γ is of index 12 in Γ , and that it is a free group with basis $\{u, v\}$ where¹

$$u = xyx^{-1}y^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \quad \text{and} \quad v = xy^{-1}x^{-1}y = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The group Γ' is therefore of type (FL), and we have

$$\text{cd}(\Gamma') = 1, \quad \chi(\Gamma') = 1 - 2 = -1.$$

We deduce from this that Γ is of type (VFL), and that

$$\text{vcd}(\Gamma) = 1, \quad \chi(\Gamma) = -\frac{1}{12}.$$

Note that $-\frac{1}{12}$ is also *the value of the Riemann zeta function at the point -1* ; we will return to this “coincidence” in no. 3.7.

- 3) More generally, every S -arithmetic group is of type (VFL), cf. no. 2.4, Th. 4.

Remark

If Γ is of type (VFL), one easily shows that the Γ -module $\mathbf{Z}$ has a free resolution of finite type (no. 1.1). It follows, for example, that the functors $M \mapsto H^q(\Gamma, M)$ commute with direct limits, and that $H^q(\Gamma, M)$ is of finite type over $\mathbf{Z}$ if M is. In addition, the groups $H^q(\Gamma, M)$ are torsion groups for $q > \text{vcd}(\Gamma)$.

Questions

Let Γ be a group of type (VFL). Suppose that Γ is torsion-free. Is it true that Γ is of type (FL)? This is true in various special cases. In the general case, I do not even know if $\chi(\Gamma)$ is an integer. The question is related to the following:

Let k be a field of characteristic zero, let V be a $k[\Gamma]$ -module of finite rank over k , and let Γ' be a normal subgroup of finite index of Γ . Let $V_i = H^i(\Gamma', V)$. The group Γ/Γ' acts on V_i ; let v_i be the character of the representation of Γ/Γ' thus defined and let $v = \sum (-1)^i v_i$. Is it true that v is a multiple of the character of the regular representation of Γ/Γ' , i.e., that $v(s) = 0$ for all $s \in \Gamma/\Gamma'$, $s \neq 1$? (This is true if Γ is of type (FL), as is easily verified.)

The above questions lead to strengthening the condition (VFL):

Definition. We say that Γ is of type (WFL) if Γ is virtually torsion-free, and if all torsion-free subgroups of finite index of Γ are of type (FL).

Proposition 13. Let Γ be a group of type (WFL) and let m be the lcm of the orders of the finite subgroups of Γ . There exists an integer $n \geq 1$ such that $m^n \cdot \chi(\Gamma) \in \mathbf{Z}$.

¹The group Γ' has a simple interpretation in terms of the modular function $f(\tau) = \eta(\tau)^2 = \Delta(\tau)^{1/12} = e^{\pi i \tau/6} \prod_{n \geq 1} (1 - e^{2\pi i n \tau})^2$; it is the set of matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbf{Z})$ such that $f\left(\frac{a\tau+b}{c\tau+d}\right) = (c\tau+d)f(\tau)$. It can be described by congruences mod. 12, cf. Hurwitz, *Math. Werke*, I, p. 35-40.

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(I do not know if one can take $n = 1$.)

It suffices to show that, if p is a prime number and if Γ does not contain any element of order p , then $\chi(\Gamma)$ is p -integral. Let Γ' be a torsion-free subgroup of finite index in Γ . Up to replacing Γ' by the intersection of its conjugates, we may assume that Γ' is normal in Γ . Let P be a Sylow p -subgroup of Γ/Γ' , and let Γ_p be its inverse image in Γ . If C is a finite subgroup of Γ_p , we have $C \cap \Gamma' = \{1\}$, which shows that C is isomorphic to a subgroup of P , hence is a p -group. Since Γ does not contain any element of order p , we thus have $C = \{1\}$, which proves that Γ_p is torsion-free. Since Γ is assumed to be of type (WFL), Γ_p is of type (FL), and $\chi(\Gamma_p)$ is an integer. But the index $(\Gamma : \Gamma_p)$ is coprime to p , and we have $\chi(\Gamma) = \chi(\Gamma_p)/(\Gamma : \Gamma_p)$. Whence the fact that $\chi(\Gamma)$ is p -integral, q.e.d.

Most of the results of the preceding sections carry over without difficulty to the "virtual" notions defined above. We will confine ourselves to two examples:

i) *Cellular actions*

We resume the notation and hypotheses of section 1.6: T is an acyclic CW-complex on which Γ acts cellularly; for every cell σ , we denote by Γ_σ the stabilizer of σ ; we choose a set Σ of representatives of the cells of T modulo Γ .

Proposition (14). (a) *If Γ is virtually torsion-free, the same holds for the Γ_σ and we have*

$$\text{vcd}(\Gamma) \leq \sup_{\sigma \in \Sigma} (\dim(\sigma) + \text{vcd}(\Gamma_\sigma)).$$

(b) *Suppose that Σ is finite, and that Γ contains a subgroup Γ' of finite index whose intersections with the conjugates of the Γ_σ are of type (FL). Then Γ and the Γ_σ are of type (VFL), and we have*

$$\chi(\Gamma) = \sum_{\sigma \in \Sigma} (-1)^{\dim(\sigma)} \chi(\Gamma_\sigma).$$

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c) Suppose that Σ is finite, that Γ is virtually torsion-free, and that the Γ_σ are of type (WFL). Then Γ is of type (WFL).

Let Γ' be a torsion-free subgroup of finite index in Γ . For each $\sigma \in \Sigma$, let I_σ be a set of representatives in Γ of the elements of $\Gamma' \backslash \Gamma/\Gamma_\sigma$. The cells $\gamma\sigma$ ($\sigma \in \Sigma, \gamma \in I_\sigma$) form a system of representatives of the cells of T modulo Γ' . The stabilizer $\Gamma'_{\gamma\sigma}$ in Γ' of such a cell $\gamma\sigma$ is the group

$$\Gamma' \cap \Gamma_{\gamma\sigma} = \Gamma' \cap (\gamma\Gamma_\sigma\gamma^{-1}) \simeq (\gamma^{-1}\Gamma'\gamma) \cap \Gamma_\sigma.$$

It is isomorphic to a torsion-free subgroup of finite index in Γ_σ . Applying Prop. 11 (a) to Γ' , we deduce

$$\begin{aligned} \text{cd}(\Gamma') &\leq \sup_{\sigma, \gamma} (\dim(\sigma) + \text{cd}(\Gamma'_{\gamma\sigma})) \\ &\leq \sup_{\sigma} (\dim(\sigma) + \text{vcd}(\Gamma_\sigma)), \end{aligned}$$

hence (a), since $\text{vcd}(\Gamma) = \text{cd}(\Gamma')$.

Suppose now that Σ is finite and that the $\Gamma'_{\gamma\sigma}$ are of type (FL). It is clear that the Γ_σ are then of type (VFL), and Prop. 11 (b), applied to Γ' , shows that Γ' is of type (FL), and thus that Γ is of type (VFL). This proves the first part of (b), as well as (c). Prop. 11 further shows that

$$\chi(\Gamma') = \sum_{\sigma \in \Sigma} \sum_{\gamma \in I_\sigma} (-1)^{\dim(\sigma)} \chi(\Gamma'_{\gamma\sigma}).$$

But $\chi(\Gamma')$ is equal to $(\Gamma : \Gamma')\chi(\Gamma)$. We are therefore reduced to proving the formula

$$(\Gamma : \Gamma')\chi(\Gamma) = \sum_{\gamma \in I_\sigma} \chi(\Gamma'_{\gamma\sigma}).$$

Since $\Gamma'_{\gamma\sigma}$ is isomorphic to the group $\Gamma_\sigma^\gamma = (\gamma^{-1}\Gamma'\gamma) \cap \Gamma_\sigma$, the right-hand side can be written

$$\sum_{\gamma \in I_\sigma} \chi(\Gamma_\sigma^\gamma) = \sum_{\gamma \in I_\sigma} \chi(\Gamma_\sigma) (\Gamma_\sigma : \Gamma_\sigma^\gamma).$$

It is therefore sufficient to prove that

$$(\Gamma : \Gamma') = \sum_{\gamma \in I_\sigma} (\Gamma_\sigma : \Gamma_\sigma^\gamma). \quad (*)$$

But $\Gamma' \setminus \Gamma$ is the disjoint union of the $\Gamma' \setminus \Gamma' \gamma \Gamma_\sigma$ for $\gamma \in I_\sigma$, and we have

$$\text{Card}(\Gamma' \setminus \Gamma' \gamma \Gamma_\sigma) = \text{Card}(\Gamma_\sigma^\gamma \setminus \Gamma_\sigma) = (\Gamma_\sigma : \Gamma_\sigma^\gamma).$$

Formula (*) follows from this.

ii) *Amalgams*

PROPOSITION 15. *Let $\Gamma = \Gamma_1 *_A \Gamma_2$ be an amalgam (cf. no. 1.3).*

(a) *Suppose Γ is virtually torsion-free. Then the same is true for Γ_1 , Γ_2 , and A , and we have*

$$\text{vcd}(\Gamma) \leq \text{Sup}(\text{vcd}(\Gamma_1), \text{vcd}(\Gamma_2), 1 + \text{vcd}(A)).$$

(b) *Suppose that Γ contains a subgroup of finite index whose intersections with the conjugates of A , Γ_1 , and Γ_2 are of type (FL).*

Then A , Γ_1 , Γ_2 , and Γ are of type (VFL), and we have

$$\chi(\Gamma) = \chi(\Gamma_1) + \chi(\Gamma_2) - \chi(A).$$

If, in addition, A , Γ_1 , and Γ_2 are of type (WFL), the same is true for Γ .

This results from Prop. 14, applied to the action of Γ on the tree T defined by the amalgam ([38], Chap. I, no. 4.1).

COROLLARY 1. *If Γ_1 and Γ_2 are finite, the group $\Gamma = \Gamma_1 *_A \Gamma_2$ is of type (WFL), and we have*

$$\text{vcd}(\Gamma) \leq 1 \quad \text{and} \quad \chi(\Gamma) = \frac{1}{\text{Card}(\Gamma_1)} + \frac{1}{\text{Card}(\Gamma_2)} - \frac{1}{\text{Card}(A)}.$$

It is known that Γ is *residually finite* (cf. for example Bourbaki, Alg. I, new ed., §7, exerc. 33). We conclude that there exists a normal subgroup of finite index of Γ which intersects Γ_1 and Γ_2 only in the identity element. We are therefore in the conditions of application of Prop. 15 (b). Whence the result.

COROLLARY 2. *Let S be a finite set of prime numbers, let $\mathbf{Z}_S$ be the subring of $\mathbf{Q}$ generated by the elements $1/p$, $p \in S$, and let $\Gamma_S = \text{SL}_2(\mathbf{Z}_S)$. The group Γ_S is of type (WFL), and we have*

$$\text{vcd}(\Gamma_S) \leq 1 + \text{Card}(S) \quad \text{and} \quad \chi(\Gamma_S) = -\frac{1}{12} \prod_{p \in S} (1 - p).$$

When $S = \emptyset$, we have $\mathbf{Z}_S = \mathbf{Z}$ and $\Gamma_S = \text{SL}_2(\mathbf{Z})$. In this case, the corollary results from Cor. 1, applied to $\Gamma_1 = \mathbf{Z}/6\mathbf{Z}$, $\Gamma_2 = \mathbf{Z}/4\mathbf{Z}$ and $A = \mathbf{Z}/2\mathbf{Z}$ (we indeed have $\frac{1}{6} + \frac{1}{4} - \frac{1}{2} = -\frac{1}{12}$). When $S \neq \emptyset$, we argue by induction on the number of elements of S . Let $p \in S$, and let $T = S - \{p\}$. According to [38], p. II-16, we can write Γ_S as an amalgam

$$\Gamma_S = \Gamma_1 *_A \Gamma_2$$

where Γ_1 and Γ_2 are isomorphic to Γ_T and where A is of index $p + 1$ in Γ_1 and Γ_2 . Cor. 1, combined with the induction hypothesis, shows that Γ_S is of type (WFL), that $\text{vcd}(\Gamma_S) \leq 1 + \text{Card}(S)$, and that

$$\begin{aligned}\chi(\Gamma_S) &= \chi(\Gamma_1) + \chi(\Gamma_2) - \chi(A) \\ &= (1 + 1 - (p + 1))\chi(\Gamma_T) = (1 - p)\chi(\Gamma_T), \\ &= (1 - p) \left(-\frac{1}{12} \right) \prod_{\ell \in T} (1 - \ell) \\ &= -\frac{1}{12} \prod_{\ell \in S} (1 - \ell), \quad \text{q.e.d.}\end{aligned}$$

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REMARKS

- 1) The inequality $\text{vcd}(\Gamma_S) \leq 1 + \text{Card}(S)$ is in fact an *equality*, cf. n° 2.5, prop. 21.
- 2) Let us set

$$\zeta_S(s) = \prod_{p \notin S} \frac{1}{1 - p^{-s}}.$$

We have:

$$\zeta_S(-1) = \zeta(-1) \prod_{p \in S} (1 - p) = -\frac{1}{12} \prod_{p \in S} (1 - p),$$

hence:

$$\chi(\Gamma_S) = \zeta_S(-1).$$

- 3) When $S = \{p\}$, we have $\Gamma_S = \text{SL}_2\left(\mathbf{Z}\left[\frac{1}{p}\right]\right)$; the Euler-Poincaré characteristic of this group is $(p - 1)/12$, its virtual cohomological dimension is equal to 2. Let Γ' be a torsion-free subgroup of finite index N in Γ_S ; we have $\chi(\Gamma') = (p - 1)N/12$; on the other hand, by Cor. 2 to Th. 5 of [39], we have $H^1(\Gamma', V) = 0$ for every $k[\Gamma']$ -module V which is of finite rank over k (k being a field of characteristic zero). From this we deduce the formula:

$$\dim_k H^0(\Gamma', V) + \dim_k H^2(\Gamma', V) = \frac{(p - 1)N}{12} \dim_k V.$$

1.9. Example: Coxeter groups.

Let S be a finite set, and let $M = (m(s, s'))$ be a Coxeter matrix of type S ([9], p. 20, Def. 4). Let us recall what this means:

- i) If s, s' belong to S , $m(s, s')$ is either an integer ≥ 1 , or $+\infty$.
- ii) The matrix M is symmetric.
- iii) We have $m(s, s') = 1$ if and only if $s = s'$.

Let W be the *Coxeter group* defined by M ([9], p. 91-92); it is generated by the elements s of S , subject to the relations

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$$(ss')^{m(s, s')} = 1 \quad \text{for} \quad (s, s') \in S \times S \quad \text{and} \quad m(s, s') < \infty.$$

In particular, we have $s^2 = 1$ for all $s \in S$.

If I is a subset of S , we denote by W_I the subgroup of W generated by I ; it is a Coxeter group (more precisely, the pair (W_I, I) is a Coxeter system, cf. [9], p. 20, th. 2). We have $W_S = W$ and $W_\emptyset = \{1\}$.

PROPOSITION 16. (a) *The group W is of type (WFL).*

(b) *Let $d = \text{Card}(S)$ and let m be the number of irreducible components of (W, S) , cf. [9], p. 21. We have $\text{vcd}(W) \leq d - m$.*

(c) *If W is infinite, we have*

$$\sum_{I \subset S} \varepsilon(I) \chi(W_I) = 0.$$

[Here, and in all that follows, we set $\varepsilon(I) = (-1)^{\text{Card}(I)}$.]

We reason by induction on d , the case $d = 0$ being trivial. Suppose $d \geq 1$. We are going to use the simplicial complex T associated by Tits to W (cf. [45] as well as [9], p. 40, exerc. 16). This complex enjoys the following properties, which characterize it up to isomorphism:

(α) W acts simplicially on T .

(β) T contains a simplex C , with vertices $(x_s)_{s \in S}$ indexed by S ; for every subset I of S , distinct from S , we denote by C_I the face of C with vertices $x_s, s \in S - I$.

(γ) If $w \in W$ and $I \subset S, I \neq S$, we have $w \cdot C_I = C_I$ if and only if $w \in W_I$, and in this case w is the identity on the face C_I .

(δ) Every simplex of T can be transformed by an element of W into one and only one face of C .

In particular, T is of dimension $d - 1$. Note that one can give a geometric interpretation of T : it is isomorphic³ to the complex obtained by

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intersecting the cone U of [9], p. 96 with a sphere Σ centered at 0 in the Euclidean space $\mathbf{R}^S$. When W is finite, we have $U = \mathbf{R}^S$ and T is homeomorphic to S_{d-1} . On the other hand:

Lemma (4). *When W is infinite, T is contractible.*

This can be deduced from the geometric interpretation of T mentioned above, taking into account that U is convex (*loc. cit.*, p. 97) and distinct from $\mathbf{R}^S$; however, one must take certain precautions, because the topology of T does not necessarily coincide with that of $\Sigma \cap U$. Here is another proof, which has the advantage of also applying to the buildings associated with Tits systems with an infinite Coxeter group ([9], p. 49, exercise 10):

We list the elements of W :

$$w_1, w_2, \dots, w_n, \dots$$

in such a way that, if $\ell(w)$ denotes the length of w ([9], p. 9, Def. 1), we have $\ell(w_n) \leq \ell(w_{n+1})$ for all $n \geq 1$. We have $w_1 = 1$. Let us denote by T_n the subcomplex of T formed by the union of the $w_i C$ for $i \leq n$; the complex T is the increasing union of the sequence of T_n and it therefore suffices to show that T_n is contractible for all n . This is done by induction on n , the case $n = 1$ being immediate since $T_1 = C$ is a simplex. Suppose $n \geq 2$. The complex T_n is obtained by adjoining the simplex $w_n C$ to T_{n-1} and everything reduces to showing that $w_n C$ is retractable onto $w_n C \cap T_{n-1}$. Let J be the set of $s \in S$ such that $\ell(w_n s) < \ell(w_n)$. Since $w_n \neq 1$, we have $J \neq \emptyset$; on the other hand, we have $J \neq S$ because otherwise w_n would be the element of greatest length in W ([9], p. 43, exercise 22) and W would be finite. Let I be a...

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subset of S distinct from S . In order for the face $w_n C_I$ of $w_n C$ to belong to T_{n-1} , it is necessary and sufficient that there exists an index $i \leq n - 1$ such that $w_n \in w_i W_I$; as $\ell(w_i) \leq \ell(w_n)$, this is only the

³More precisely, one defines a canonical bijection from T onto $\Sigma \cap U$ which is continuous and respects the simplicial structures of these two spaces; however, this is not in general a homeomorphism.

case if w_n is not the element of minimal length of the coset $w_n W_I$; according to [9], p. 37, Exercise 3, this is equivalent to saying that $\ell(w_n s) < \ell(w_n)$ for at least one $s \in I$, i.e., that I meets J . We thus see that $w_n C \cap T_{n-1}$ is the *union of the faces* $w_n C_{\{j\}}$ for $j \in J$. Since these faces have codimension 1, and J is distinct from both $\emptyset$ and S , we indeed deduce that $w_n C$ retracts onto $w_n C \cap T_{n-1}$, which completes the proof of the lemma.

Let us now return to the proof of Prop. 16. If W is finite, there is nothing to prove. Suppose then that W is infinite. Since W is isomorphic to a finitely generated subgroup of $\mathbf{GL}_d(\mathbb{R})$ (cf. [9], p. 93), it is virtually torsion-free (see also [9], p. 131, Exercise 9). In view of Lemma 4, we can apply Prop. 14 to the complex T , by choosing for Σ the set of faces C_I ($I \neq S$) of C . The induction hypothesis shows that the W_I are of type (WFL), and therefore the same holds for W . Furthermore, we have

$$\begin{aligned} \text{vcd}(W) &\leq \sup_{I \neq S} (\dim(C_I) + \text{vcd}(W_I)) \\ &\leq \sup_{I \neq S} (d - 1 - \text{Card}(I) + (\text{Card}(I) - m_I)), \end{aligned}$$

where m_I denotes the number of irreducible components of W_I . We deduce

$$\text{vcd}(W) \leq d - 1,$$

which proves (b) when W is irreducible; the general case reduces to this one by decomposing W into irreducible factors and applying, for example, Prop. 6. Finally, Prop. 14 shows that we have

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$$\begin{aligned} \chi(W) &= \sum_{I \neq S} (-1)^{\dim(C_I)} \chi(W_I) \\ &= -\varepsilon(S) \sum_{I \neq S} \varepsilon(I) \chi(W_I), \end{aligned}$$

whence, by regrouping,

$$\sum_{I \subseteq S} \varepsilon(I) \chi(W_I) = 0, \quad \text{q.e.d.}$$

REMARKS

1) The action of W on T gives other information on the cohomology of W with values in a W -module M :

For any integer $q \geq 0$, let us define a “simplicial sheaf” $\mathcal{H}^q(M)$ on the simplex C by associating to each face C_I of C the group $H^q(W_I, M)$, and to each inclusion $C_I \subset C_J$ the homomorphism

$$\text{Res}: H^q(W_I, M) \rightarrow H^q(W_J, M).$$

Let us denote by $H^\bullet(C, \mathcal{H}^q(M))$ the cohomology of the simplex C with values in the sheaf $\mathcal{H}^q(M)$. If W is infinite, the fact that T is acyclic implies the existence of a *spectral sequence*, converging to $H^\bullet(W, M)$, whose term $E_2^{p,q}$ is equal to $H^p(C, \mathcal{H}^q(M))$; the case where W is finite is analogous, but a bit more complicated. In both cases, we obtain close relations between $H^\bullet(W, M)$ and the $H^\bullet(W_I, M)$ for $I \neq S$; one should be able to use them to study $H^\bullet(W, M)$ in more detail.

2) The formula

$$\sum_{I \subseteq S} \varepsilon(I) \chi(W_I) = 0 \quad (\text{for } W \text{ infinite})$$

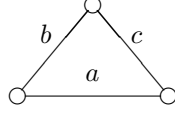
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allows one to calculate $\chi(W)$ by induction on $d = \text{Card}(S)$, taking into account that $\chi(W_I) = 1/\text{Card}(W_I)$ when W_I is finite. Here are a few examples:

i) (Schwarz *triangle* groups). We take for the Coxeter graph a triangle

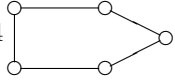


and we denote by a, b, c the $m(s, s')$ corresponding to its sides. We assume that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq 1$, which is equivalent to saying that W is infinite ([9], p. 130, exercise 4). The W_I corresponding to $\text{Card}(I) = 2$ are of order $2a, 2b, 2c$; those corresponding to $\text{Card}(I) = 1$ are of order 2; we thus find:

$$\begin{aligned}\chi(W) &= (1/2a + 1/2b + 1/2c) - 3/2 + 1 \\ &= \frac{1}{2} \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} - 1 \right).\end{aligned}$$

ii) Graph $\circ - \circ - \circ - \circ - \circ$ ⁵; we find $\chi(W) = 1/2^6 3^2 5^2$.

iii) Graph $\circ - \circ - \circ - \circ - \circ$ ⁴ ⁵; we find $\chi(W) = 17/2^7 3^2 5^2$.

iv) Graph ; we find $\chi(W) = 11/2^7 3^2 5$.

[The graphs ii), iii), iv) above are of *compact hyperbolic type* (cf. Vinberg [46] as well as [9], p. 133, exercise 15); the same holds for graph i) if a, b, c are finite and $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} < 1$. More generally, when W is of hyperbolic type, we have $\chi(W) = 0$ if d is even, $\chi(W) > 0$ if $d \equiv 1 \pmod{4}$ and $\chi(W) < 0$ if $d \equiv 3 \pmod{4}$; moreover, $\text{vcd}(W) = d - 1$ if W is of compact type, and $\text{vcd}(W) = d - 2$ otherwise.]

The following proposition gives another way to calculate $\chi(W)$:

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PROPOSITION 17. *Let us denote by $W(t)$ the formal series $\sum_{w \in W} t^{\ell(w)}$. Then $W(t)$ is a rational function of t , and we have*

$$\chi(W) = 1/W(1).$$

The fact that $W(t)$ is a rational function is proved in [9], p. 45, exerc. 26. This gives a meaning to the expression $1/W(1)$. To prove that $1/W(1)$ is equal to $\chi(W)$, we proceed by induction on $d = \text{Card}(S)$, the case $d \leq 1$ being trivial. If W is finite, we have $W(1) = \text{Card}(W)$, and the formula is obvious. Suppose W is infinite. The induction hypothesis, combined with Prop. 16, shows that

$$\varepsilon(S)\chi(W) = - \sum_{I \neq S} \varepsilon(I)/W_I(1).$$

But, according to [9], *loc. cit.*, f), we have $\sum_{I \subset S} \varepsilon(I)/W_I(t) = 0$, whence, by setting $t = 1$, the equation

$$\varepsilon(S)/W(1) = - \sum_{I \neq S} \varepsilon(I)/W_I(1).$$

By comparing, we indeed obtain $\chi(W) = 1/W(1)$, q.e.d.

REMARK

When W is finite, or “Euclidean”, one can give simple explicit formulas for $W(t)$, cf. [9], pp. 230-231, exerc. 10. I know of nothing of this kind in the general case. In particular, I do not know what the poles of $W(t)$ are (its zeros are roots of unity distinct from 1, cf. [9], p. 45, exerc. 26). When W is of irreducible Euclidean type, or compact hyperbolic, one easily proves the identity

$$W(t^{-1}) = -\varepsilon(S)W(t),$$

cf. no. 3.4, Prop. 26 (d); this gives a symmetry property of the poles in question, and shows that they are units in an algebraic number field.

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§2. DISCRETE SUBGROUPS OF PRODUCTS OF LIE GROUPS

In this §, Γ is a discrete subgroup of a finite product $G = \prod G_a$ of Lie groups (real or ultrametric), and we propose to relate the cohomology of Γ to the properties of the G_a . We shall only concern ourselves with the invariants $\text{cd}(\Gamma)$, $\text{vcd}(\Gamma)$ as well as the properties (FL), (VFL) and (WFL) defined in §1; for $\chi(\Gamma)$, see §3.

2.1. Review of the real case.

Let G be a real Lie group having a finite number of connected components, and let K be a maximal compact subgroup of G . We shall set

$$d(G) = \dim(G/K) = \dim(G) - \dim(K).$$

It is known that the homogeneous space $T = G/K$ is diffeomorphic to the Euclidean space $\mathbb{R}^{d(G)}$; in particular, T is contractible. Moreover, G acts *properly* on T .

Let Γ be a discrete subgroup of G . If $x \in T$, the stabilizer Γ_x of x in Γ is compact and discrete, hence *finite*. If Γ is torsion-free, it follows that Γ acts *freely* on T , so that T is a universal covering of the quotient manifold $X_\Gamma = \Gamma \backslash T = \Gamma \backslash G/K$.

The space X_Γ is a $K(\Gamma, 1)$ space, cf. no. 1.5. Moreover, since X_Γ is a differentiable manifold, it can be *triangulated*.²

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PROPOSITION 18. *Let Γ be a torsion-free discrete subgroup of G .*

- (a) *If G/Γ is compact, Γ is of type (FL), and $\text{cd}(\Gamma) = d(G)$.*
- (b) *If G/Γ is not compact, we have $\text{cd}(\Gamma) \leq d(G) - 1$.*

This follows from what precedes, and from Prop. 9 of no. 1.5. For (a), we must furthermore remark that G/Γ is compact if and only if the manifold X_Γ is compact; for (b), we use the following well-known lemma (a consequence of Poincaré duality):

LEMMA 5. *Let X be a non-compact connected topological manifold, and let M be a local system of coefficients on X . We have $H^n(X, M) = 0$ for all $n \geq \dim(X)$.*

Proposition 18 obviously implies:

COROLLARY. *Let Γ be a discrete subgroup of G . Suppose Γ is virtually torsion-free. Then we have $\text{vcd}(\Gamma) \leq d(G)$ and there is equality if and only if G/Γ is compact; in this latter case, Γ is of type (WFL).*

REMARK

Let us return to the case where Γ is torsion-free. We say that $X_\Gamma = \Gamma \backslash T$ is *bordable* if there exists a compact C^∞ manifold with boundary Y_Γ whose interior is isomorphic to X_Γ . Since the spaces X_Γ and Y_Γ have the same homotopy type, Y_Γ is also a $K(\Gamma, 1)$ space. But any differentiable manifold with boundary can be triangulated. We deduce that, if X_Γ is bordable, the group Γ is of type (FL). This applies in particular to the case where G is semisimple and Γ is arithmetic (Raghunathan [33] – see also no. 2.4, proof of Thm. 4). Let us mention in this connection:

QUESTION

Let μ be a right Haar measure on G . Suppose that $\mu(G/\Gamma)$ is finite. Is it true that X_Γ is bordable? This is true when G is a semisimple group of real rank 1, by Garland–Raghunathan [17].

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²For the triangulation of differentiable manifolds (possibly with boundary), see J. R. Munkres, *Elementary Differential Topology*, Ann. of Math. Studies no. 54, §10. [In fact, in most applications, a *cell decomposition* suffices, and this can be obtained at less cost, thanks to Morse theory.]

2.2. Review of the ultrametric case

Let k be a *local field*; in all that follows, we shall mean by this a field *complete* for a discrete valuation v , with *perfect* residue field.⁵ We assume v is normalized such that $v(k^*) = \mathbb{Z}$; we denote by O_v (resp. $\mathfrak{m}_v$, resp. $\bar{k}$) the ring of v (resp. its maximal ideal, resp. its residue field); we have $\bar{k} = O_v/\mathfrak{m}_v$. The most important case for what follows is the one where k is *locally compact*, i.e., where $\bar{k}$ is *finite*.

Let L now be a *linear algebraic group* over k . Let G denote the group $L(k)$ of its points with coefficients in k ; this is a topological group. Since L is an affine variety, the notion of a bounded subset of G has an obvious meaning (it is a subset which is bounded in *an* affine embedding, or in *all*, which comes to the same thing). When k is locally compact, the same holds for G , and “bounded” is equivalent to “relatively compact”.

Suppose that L is *connected reductive*, and let $d(G)$ denote its rank over k in the sense of Borel-Tits [8], i.e., the dimension of a maximal split torus of L . According to Bruhat and Tits, one can associate to L a CW-complex T , on which G acts cellularly, and which plays a role analogous to that of the homogeneous space G/K of the preceding section. We call T *the building* of G . It enjoys the following properties:

- i) T is *contractible*.
- ii) We have $\dim(T) = d(G)$.
- iii) For every cell σ of T , the stabilizer G_σ of σ in G is an *open bounded subgroup* of G .
- iv) Every bounded subgroup of G is contained in a G_σ .
- v) The cells of T are *finite in number* modulo the action of G .

We begin with two special cases:

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- a) L is the *multiplicative group* G_m (so that $G = k^*$).

We take for T the line $\mathbf{R}$, equipped with the triangulation defined by the *integer points*, and by the intervals $[n, n+1]$, $n \in \mathbf{Z}$. If $g \in k^*$ is an element of G , we make g act on T by translation of amplitude $v(g)$. The G_σ are all equal to the *group of units* of k^* , which is the largest bounded subgroup of G . It is clear that properties i) to v) above are satisfied.

- b) L is *simple over* k .

This is the crucial case, treated by Bruhat-Tits (cf. [11], [12]). Let $\bar{L}$ be the universal covering (in the algebraic sense) of L . Bruhat and Tits begin by equipping the group $\bar{L}(k)$ with a *Tits system* structure (cf. [9], chap. IV, §2); the Weyl group W of this system is an infinite irreducible Euclidean Coxeter group of rank equal to $d(G) + 1$. [When L is *split*, the Tits system in question had been defined by Iwahori and Matsumoto [23]; the group W is then simply the *affine Weyl group* of the root system of L . The quasi-split case is due to Hijikata [20]. Bruhat and Tits show how the general case can be reduced to these special cases by a suitable “Galois descent”.]

Once the Tits system of $\bar{L}(k)$ is obtained, one defines T as the *building associated with this system* (cf. [12], I, §2 or [9], p. 49, exerc. 10); it is a *simplicial complex*; its faces correspond to the “parahoric” subgroups of $\bar{L}(k)$. The group $\bar{L}(k)$ acts simplicially on T , with a fundamental domain being a simplex of dimension $d(G)$. More generally, the automorphism group of $\bar{L}$ acts naturally on T ; since L maps in an obvious way into the *adjoint group* of $\bar{L}$, we indeed deduce an action of $G = L(k)$ on T . Properties ii) and v) are immediate; iii) results from the fact that a “parahoric” group is open and bounded; i) and iv) are general properties of buildings with a Coxeter group of Euclidean type, cf. [10], [12].

EXAMPLE

Let us take for L the group SL_n , $n \geq 2$, which is simply connected.

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⁵I do not know if this hypothesis is indispensable.

The building T is interpreted in the following way: let R be the set of *lattices* of k^n (free sub- O_v -modules of rank n), and let $\bar{R}$ be the quotient of R by k^* acting by homotheties. The set $\bar{R}$ is the *set of vertices* of T . A finite subset of $\bar{R}$ is a *face* of T if one can represent its elements by lattices $L_1, \dots, L_p$ such that

$$L_1 \supset L_2 \supset \dots \supset L_p \supset \mathfrak{m}_v L_1.$$

The stabilizer of such a face is simply the set of $g \in G$ such that $g(L_i) = L_i$ for $i = 1, \dots, p$.

When $n = 2$, we recover *the tree of* SL_2 , cf. [38], chap. II.

c) Construction of T : general case.

Let C be the center of L , and let $L/C = \prod_{\alpha \in A} L_\alpha$ be the decomposition of the semisimple group L/C into a product of simple groups. For every $\alpha \in A$, let T_α be the building of L_α , cf. b). The group $G = L(k)$ acts simplicially on T_α , via the homomorphism $L \rightarrow L_\alpha$.

Let, on the other hand, Φ be a *basis* of the abelian group $\mathrm{Hom}_k(L, \mathbf{G}_m)$. For every $\phi \in \Phi$, let T_ϕ be the building of $\mathbf{G}_m$, cf. a), and we let $L(k)$ act on T_ϕ by means of ϕ .

We then set

$$T = \prod_{\alpha \in A} T_\alpha \times \prod_{\phi \in \Phi} T_\phi.$$

This is a *polysimplicial complex* in the sense of [12], I, 1.1; in particular, it is a CW-complex. The group G acts in a natural way on T . Properties i) to v) are verified by noting that L is isogenous to a product

$$L_1 = L' \times \prod_{\alpha \in A} L_\alpha \times \prod_{\phi \in \Phi} \mathbf{G}_m,$$

where L' is of rank 0; the group $L'(k)$ is then *bounded* (cf. [11], Cor. to

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Thm. 4), and one easily reduces in this way to cases a) and b) considered above; we leave the details of this reduction to the reader (especially since the case where L is *isomorphic* to L_1 would suffice for us in the sequel).

REMARKS

1) When k is locally compact, the complex T is locally finite, hence also locally compact; the stabilizers G_σ of the cells σ of T are *compact open subgroups* of G .

2) The complex structure of T depends on the choice of the basis Φ of $\mathrm{Hom}_k(L, \mathbf{G}_m)$; this is harmless for applications.

Applications to the cohomology of groups

Let Γ be a group, and let i be a homomorphism from Γ into the group $G = L(k)$. If U is a subgroup of G , we set $\Gamma_U = i^{-1}(U)$. We denote by $\mathfrak{B}$ the set of *bounded open subgroups* of G .

PROPOSITION 19. *We have $\mathrm{cd}(\Gamma) \leq \mathrm{d}(G) + \sup_{U \in \mathfrak{B}} \mathrm{cd}(\Gamma_U)$.*

We make Γ operate on T by means of $i : \Gamma \rightarrow G$. The stabilizers Γ_σ of the cells σ of T are of the form Γ_U , with $U \in \mathfrak{B}$, by property iii) of the building T . Prop. 19 follows from this, and from Prop. 11 (a) of no. 1.6.

PROPOSITION 20. *Suppose that, for all $U \in \mathfrak{B}$, Γ_U is of type (FL), and that $i(\Gamma) \backslash G/U$ is finite. Then Γ is of type (FL).*

Let $\mathfrak{S}$ be the set of cells of T . By properties iii) and v) of the building T , the G -set $\mathfrak{S}$ is the disjoint union of a finite number of homogeneous spaces of the form G/U , with $U \in \mathfrak{B}$. The quotient of $\mathfrak{S}$ by Γ is therefore a finite union of sets of the form $i(\Gamma) \backslash G/U$; by the hypothesis made on Γ , it is finite. The proposition follows from this, and from Prop. 11 (b) of no. 1.6.

REMARKS

1) In the above statements, one can replace $\mathfrak{B}$ by the set

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$\mathfrak{B}'$ of the stabilizers of the cells of T (when L is simple and simply connected, $\mathfrak{B}'$ is the set of *parahoric subgroups* of G , in the sense of Bruhat-Tits).

2) Suppose k is locally compact. Then $\mathfrak{B}$ is the set of *open compact subgroups* of G ; if U, U' belong to $\mathfrak{B}$, $U \cap U'$ is of finite index in U and U' ; one deduces an analogous property for the Γ_U . Taking into account Theorem 1 of no. 1.7, we see that, if Γ is torsion-free, the Γ_U have the *same cohomological dimension*, which simplifies the statement of Prop. 19; similarly, in the case of Prop. 20, we can restrict ourselves to subgroups U which are *maximal compact* in G .

QUESTION

Let us take for k the p -adic field $\mathbf{Q}_p$, and let $H_{\text{cont}}^m(G, \mathbf{Q}_p)$ (resp. $H_{\text{ana}}^m(G, \mathbf{Q}_p)$) be the m -th cohomology group of the complex of *continuous* (resp. *analytic*) cochains of G with values in $\mathbf{Q}_p$. When $d(G) = 0$, the group G is *compact*, and a theorem of Lazard ([27], Chap. V, §2) shows that

$$H_{\text{cont}}^m(G, \mathbf{Q}_p) \simeq H_{\text{ana}}^m(G, \mathbf{Q}_p) \simeq H^m(\mathfrak{g}, \mathbf{Q}_p),$$

where $\mathfrak{g}$ denotes the *Lie algebra* of G . Do we have an analogous result in the general case? It should be possible to use the action of G on the building T to relate the cohomology groups in question with those of the open compact subgroups of G , which are amenable to Lazard's theory. The first case to consider is that of $\text{SL}_2(\mathbf{Q}_p)$, which is the amalgamated sum of two copies of $\text{SL}_2(\mathbf{Z}_p)$.

2.3. Passage to products

In this section, $(G_\alpha)_{\alpha \in A}$ denotes a finite family of Lie groups. We assume that each G_α is of one of the types considered in sections 2.1 and 2.2, in other words:

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- (i) a real Lie group having a finite number of connected components;
- (ii) the group $L_\alpha(k_\alpha)$ of k_α -points of a connected reductive algebraic group L_α over a local field k_α .

In case (i), we denote by T_α the quotient of G_α by one of its maximal compact subgroups; in case (ii), we denote by T_α the building of G_α . We set $d(G_\alpha) = \dim(T_\alpha)$; in case (ii), this is also the rank of G_α over k_α . Finally, we set

$$G = \prod_{\alpha \in A} G_\alpha, \quad T = \prod_{\alpha \in A} T_\alpha \quad \text{and} \quad d(G) = \dim(T) = \sum_{\alpha \in A} d(G_\alpha).$$

The group G acts on T .

A subset H of G is said to be *bounded* if its projections onto the G_α of type (i) (resp. of type (ii)) are relatively compact (resp. are bounded). We say that H is *strongly discrete*³ if every bounded subset of H is finite. When the k_α are locally compact, “bounded” is equivalent to “relatively compact” and “strongly discrete” is equivalent to “discrete”.

Theorem (2). *Let Γ be a subgroup of G which is torsion-free and strongly discrete. We then have $\text{cd}(\Gamma) \leq d(G)$.*

³When the product $G = \prod G_\alpha$ is reduced to a single factor of type (ii), saying that H is strongly discrete is equivalent to saying that H is a zero-dimensional subvariety of G , in the sense of Tate's *rigid analytic geometry*. Let us point out in this connection the following question: if Γ is a strongly discrete subgroup of G , can one define in a reasonable way a quotient rigid variety G/Γ ? This is the case if G is a torus, according to Tate, Morikawa, and Raynaud; I do not know what the situation is when $G = \text{SL}_2(\mathbf{Q}_p)$, for example.

We argue by induction on the number of factors G_α that are of type (ii). When there are none, G is of type (i), and the theorem follows from Prop. 18. Suppose then that there is one, say G_α , and let us denote by H the product of the G_β , $\beta \neq \alpha$. Let U be a bounded open subgroup of G_α ,

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and let Γ_U be the intersection of Γ with $H \times U$. The kernel of the projection

$$\text{pr}_1 : \Gamma \rightarrow H$$

is contained in $\{1\} \times U$, hence bounded. Since Γ is strongly discrete, this kernel is finite, and, since Γ is torsion-free, it is reduced to $\{1\}$. We can therefore identify Γ_U by means of pr_1 with a subgroup of H , and it is immediate that this subgroup is strongly discrete in H . By the induction hypothesis, we thus have $\text{cd}(\Gamma_U) \leq d(H)$. Whence, by applying Prop. 19,

$$\begin{aligned} \text{cd}(\Gamma) &\leq d(G_\alpha) + \sup \text{cd}(\Gamma_U) \\ &\leq d(G_\alpha) + d(H) = d(G), \end{aligned}$$

which proves the theorem.

COROLLARY. *If $d(G) = 1$, every strongly discrete torsion-free subgroup of G is a free group.*

Indeed, we then have $d(\Gamma) \leq 1$, and we apply the Stallings-Swan theorem, cf. 1.4, c).

EXAMPLE. When $G = \mathbf{PGL}_2(k)$, where k is a local field, the above corollary yields a theorem of Ihara ([22], see also [38], p. II-20).

Suppose now that the G_α are *locally compact*, in which case the same holds for G .

THEOREM 3. *Let Γ be a discrete torsion-free subgroup of G such that G/Γ is compact. Then Γ is of type (FL).*

The reasoning is the same as for Thm. 2: we proceed by induction on the number of factors G_α of type (ii). When there are none, G is of type (i) and the theorem results from Prop. 18. When there is one, say G_α , we denote by H the product of the G_β , $\beta \neq \alpha$. If U is an open compact subgroup of G_α , the group $\Gamma_U = \Gamma \cap (H \times U)$ is identified with

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... a discrete subgroup of H . Let the subgroup $H \times U$ act on the right on the homogeneous space $\Gamma \backslash G$; the orbits of $H \times U$ are open, and since $\Gamma \backslash G$ is compact, we see that they are closed (hence compact), and finite in number. Since $\Gamma \backslash G / (H \times U)$ is identified with $\text{pr}_a(\Gamma) \backslash G_a / U$, we deduce that this latter set is *finite*. In view of Prop. 20, we are thus reduced to proving that the Γ_U are of type (FL). But the quotient $(H \times U) / \Gamma_U$ is one of the orbits of $H \times U$ acting on $\Gamma \backslash G$, so it is compact by the preceding. Since U is compact, we deduce from this that H / Γ_U is compact. The induction hypothesis then shows that Γ_U is of type (FL), which completes the proof.

COROLLARY. *Let Γ be a discrete subgroup of G , which is virtually torsion-free. We then have $\text{vcd}(\Gamma) \leq d(G)$. If in addition G/Γ is compact, Γ is of type (WFL).*

This follows from the preceding theorem, applied to the torsion-free subgroups of finite index in Γ .

VARIANT

One might consider proving Th. 3 by making Γ act on the product T of the T_a . The action of Γ is proper and free, and T is contractible, so that T/Γ is a $K(\Gamma, 1)$ space. Furthermore, since G/Γ is compact, the same holds for T/Γ . To pass from this to the fact that Γ is of type (FL), it would therefore suffice to show that T/Γ admits a cellular decomposition. This is probably feasible (by “stratifying” T/Γ using manifolds with corners), but in any case less simple than the induction argument used above.

REMARKS

- 1) If Γ satisfies the hypotheses of Th. 3, one can show that $\text{cd}(\Gamma) = d(G)$.
- 2) It would be interesting to establish “vanishing theorems” analogous to those of the real case. Let us take for example the case where $G = L(k)$, L being a...

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simple algebraic group over a local field k , a finite extension of $\mathbf{Q}_p$; let Γ be a discrete subgroup of G such that G/Γ is compact; *is it true that $H^m(\Gamma, k) = 0$ for $0 < m < d(G)$?* (Results in this direction have recently been obtained by H. Garland.) The same question arises for other representations ρ of G ; the case $m = 1$, $\rho = \text{Ad}$, is particularly interesting, because it is linked to the *rigidity* of Γ .

2.4. *S*-arithmetic groups.

Let us first recall the definition of these groups:

Let k be a *global* field, i.e., an algebraic number field (finite extension of $\mathbf{Q}$) or a function field of one variable over a finite field. Let Σ be the set of places of k and let Σ^∞ be the subset of Σ consisting of Archimedean places. If v belongs to Σ , we will denote by k_v the completion of k for the topology defined by v ; the field k_v is isomorphic to $\mathbf{R}$ or $\mathbf{C}$ if $v \in \Sigma^\infty$; otherwise, k_v is a locally compact local field.

Let S be a non-empty finite subset of Σ containing Σ^∞ . We will denote by O_S the subring of k consisting of elements whose valuation is ≥ 0 at all places not belonging to S . This is a Dedekind ring, whose maximal ideals correspond to the elements of $\Sigma - S$.

Let L be a linear algebraic group over the field k ; let us choose an embedding of L into a linear group GL_n , and let us denote by Γ_S the group $L(k) \cap \text{GL}_n(O_S)$. A subgroup Γ of $L(k)$ is said to be *S*-arithmetic if it is commensurable with Γ_S , i.e., if $\Gamma \cap \Gamma_S$ is of finite index in Γ and in Γ_S ; this property does not depend on the choice of the embedding $L \rightarrow \text{GL}_n$.

For all $v \in S$, let us denote by G_v the group $L(k_v)$ of points of L with values in k_v ; this is a Lie group over k_v . Let us set

$$G = \prod_{v \in S} G_v.$$

The group G is locally compact; if Γ is an *S*-arith-

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...tic of $L(k)$, we see immediately that Γ is a *discrete subgroup* of G .

Let us now restrict ourselves to the case where L is *connected reductive*. The definitions and results of the preceding section then apply to the groups G_v and to their product G . In particular, we define $d(G_v)$ as follows:

- if $v \notin \Sigma^\infty$, $d(G_v)$ is the rank of L over k_v ;
- if $v \in \Sigma^\infty$, $d(G_v)$ is the (real) dimension of the quotient of G_v by a maximal compact subgroup.

We set

$$d(G) = \sum_{v \in S} d(G_v).$$

THEOREM 4. *Let Γ be an *S*-arithmetic subgroup of $L(k)$, where L is connected reductive.*

- (a) *Suppose that k is a number field. Then Γ is of type (WFL) and we have $\text{vcd}(\Gamma) \leq d(G)$; if the rank of L over k is ≥ 1 (i.e., if G/Γ is not compact, cf. [6], 8.12), we have $\text{vcd}(\Gamma) \leq d(G) - 1$.*
- (b) *Suppose that k is a function field over a finite field, and that the rank of L over k is 0. Then Γ is of type (WFL) and we have $\text{vcd}(\Gamma) \leq d(G)$.*

Case (a)

We know that Γ is virtually torsion-free (cf. for example [7], 17.4). We can therefore assume that Γ is *torsion-free*. We then distinguish three cases:

(a₁) *The rank of L over k is 0*

The quotient G/Γ is compact ([6], *loc. cit.*) and Th. 3 shows that Γ is of type (FL); on the other hand, Th. 2 shows that $\text{cd}(\Gamma) \leq d(G)$.

(a₂) *The rank of L over k is ≥ 1 , and $S = \Sigma^\infty$*

This case is deduced from Raghunathan [33]. More precisely, we first reduce, by restriction of scalars, to the case where $k = \mathbf{Q}$. We have

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then $G = L(\mathbf{R})$ and Γ is an arithmetic subgroup of $L(\mathbf{Q})$. Let C be the identity component of the center of L ; since L is reductive, it is a torus. Let D be the largest split sub-torus of C , and let $L' = L/D$. The group $\Gamma \cap D(\mathbf{Q})$ is an arithmetic subgroup of $D(\mathbf{Q})$, hence finite, and since Γ is torsion-free, we have $\Gamma \cap D(\mathbf{Q}) = \{1\}$; thus Γ identifies with a subgroup of $L'(\mathbf{Q})$ and it is easy to see that this subgroup is arithmetic. We are thus reduced to the case where $D = \{1\}$, i.e., where the rank of C over $\mathbf{Q}$ is zero. The quotient of $C(\mathbf{R})$ by $\Gamma \cap C(\mathbf{Q})$ is then compact. On the other hand, let $M = L/C$; this is a semisimple group, and the image Γ_M of Γ in $M(\mathbf{Q})$ is an arithmetic subgroup (which may *a priori* have torsion). Now let K be a maximal compact subgroup of $G = L(\mathbf{R})$, and let K_M be a maximal compact subgroup of $M(\mathbf{R})$ containing the image of K under the projection homomorphism $p : G \rightarrow M(\mathbf{R})$. According to [33], there exists a map $f : M(\mathbf{R}) \rightarrow \mathbf{R}_+$, which is of class C^∞ and satisfies the following properties:

- (i) f is right invariant under Γ_M and left invariant under K_M ;
- (ii) the map $f_\Gamma : M(\mathbf{R})/\Gamma_M \rightarrow \mathbf{R}_+$ obtained from f by passing to the quotient is *proper* (i.e., $f_\Gamma(x)$ tends to infinity with x);
- (iii) there exists a compact subset of $M(\mathbf{R})/\Gamma_M$ outside of which f_Γ has no critical point (i.e., df_Γ does not vanish).

By composing f with p , we obtain a function $F : G \rightarrow \mathbf{R}_+$. This function F enjoys properties (i), (ii), (iii) above relative to the groups K and Γ ; this is immediate for (i), and for (ii) and (iii) it follows from the fact that $G/\Gamma \rightarrow M(\mathbf{R})/\Gamma_M$ is *proper* (since $C(\mathbf{R})/(\Gamma \cap C(\mathbf{Q}))$ is compact). We can now apply to F the argument of Raghunathan [33]: let $X_\Gamma = K \backslash G/\Gamma$ and let $F_X : X_\Gamma \rightarrow \mathbf{R}_+$ be the function defined by F . Since Γ is torsion-free, X_Γ is a differentiable manifold, and F_X enjoys properties (ii) and (iii) above. If c is a real number

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sufficiently large, a standard argument shows that X_Γ is diffeomorphic to the interior of the manifold with boundary $Y_\Gamma = \{x \mid F_\Gamma(x) \leq c\}$, which is compact. It follows that X_Γ is *bordable*, cf. no. 2.1; the group Γ is therefore of type (FL) and we have $\text{cd}(\Gamma) \leq d(G) - 1$ by Prop. 18.

(a₃) *The rank of L over k is ≥ 1 , and $S \neq \Sigma^\infty$*

We proceed by induction on the number of elements of $S - \Sigma^\infty$. Let $v \in S - \Sigma^\infty$ and let us set $S_v = S - \{v\}$. If U is an open compact subgroup of G_v , let us denote by Γ_U the subgroup of Γ consisting of those elements whose image in G_v belongs to U ; it is immediate that Γ_U is an S_v -*arithmetic subgroup of $L(k)$* . By the induction hypothesis (resp. case (a₂) if $S_v = \Sigma^\infty$), the group Γ_U is of type (FL) and we have $\text{cd}(\Gamma_U) < d(G) - d(G_v)$. On the other hand, the set $\Gamma \backslash G_v/U$ is *finite*, by virtue of a result of Borel ([6], p. 28, formula (1)). We can therefore apply Props. 19 and 20, and we indeed deduce that Γ is of type (FL) and that $\text{cd}(\Gamma) < d(G)$.

Case (b)

Since the rank of L over k is 0, the quotient G/Γ is *compact*: this follows from a recent theorem of Harder (cf. [18], Kor. 2.2.7 for the analogous statement in the “adelic” case, a statement which easily implies the one we need). Let us then make Γ act on the CW-complex T which is the product of the buildings T_v of G_v , $v \in S$. Since G/Γ is compact, the cells of T are finite in number modulo the action

of Γ ; moreover, the stabilizers Γ_σ of the cells σ of T are the *finite subgroups* of Γ . We deduce that these are *finite in number*, up to conjugation in Γ . Since Γ is residually finite, there exists a subgroup of finite index in Γ which intersects the Γ_σ only in the identity element, and which is therefore *torsion-free*. This already shows that Γ is virtually without

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torsion. Suppose now that Γ is torsion-free. Then Γ operates freely on T , and the quotient complex T/Γ is finite. Hence Γ is of type (FL) and $\text{cd}(\Gamma) \leq \text{d}(\text{G})$, which proves (b). [The fact that Γ is isomorphic to the fundamental group of the finite complex T/Γ implies that Γ is a **finitely presented** group, a result which can also be proved by the methods of Behr [2], [3].]

COMPLEMENTS

1) In (b), we may restrict ourselves to assuming that the **semi-simple rank of* L *is zero**; the proof is analogous. On the other hand, when the semi-simple rank of L is ≥ 1 , the group L contains a subgroup isomorphic to the additive group G_a ; if p is the characteristic of k , we deduce that Γ contains an infinite direct sum of cyclic groups of order p . Thus, Γ is not virtually torsion-free, nor **a fortiori** of type (WFL). To obtain positive results, one must therefore “neglect p ”. More precisely, we show that Γ contains a subgroup of finite index Γ' which is **free of p' -torsion** (i.e., every element of Γ' is of infinite order or of order a power of p), and that we have $\text{cd}_A(\Gamma') \leq \text{d}(\text{G})$ for any commutative ring A in which p is invertible, for example $\mathbb{Z}[\frac{1}{p}]$.

2) Suppose that k is a number field, let P be a parabolic subgroup of L , and let Γ be an S -arithmetic subgroup of $P(k)$ which is **neat** ([7], 17.1). Then Γ is **of type (FL)**. Let us briefly outline the proof. Let N be the unipotent radical of P , and let $M = P/N$; the group M is **reductive**. If $S = \Sigma^\infty$, the group $\Gamma_N = \Gamma \cap N(k)$ is a finitely generated torsion-free nilpotent group, hence of type (FL), cf. no. 1.4, e); on the other hand, Γ/Γ_N is isomorphic to an arithmetic subgroup of $M(k)$, and has no torsion because Γ is neat; by Th. 4, it is a group of type (FL), and Prop. 6 shows that the same holds for Γ . When $S \neq \Sigma^\infty$, we proceed by induction on $\text{Card}(S - \Sigma^\infty)$, as in the proof of Th. 4, (a). The only thing to verify is that, if U is an open subgroup of G_v , with $v \in S - \Sigma^\infty$, the set

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$\Gamma \backslash G_v/U$ is *finite*. Now, it is known that $G_v/P(k_v)$ is compact; it follows that it is sufficient to verify the finiteness of $\Gamma \backslash P(k_v)/U_P$, where U_P is an open subgroup of $P(k_v)$; this is easily deduced from the corresponding result for $M(k_v)$, proved by Borel ([6], *loc. cit.*).

THEOREM 5. *Let Γ be a finitely generated torsion-free subgroup of $\text{GL}_n(k)$. We have $\text{cd}(\Gamma) < \infty$.*

Since Γ is finitely generated, we can choose S large enough so that Γ is contained in the group $\Gamma_S = \text{GL}_n(O_S)$. Let us then distinguish two cases:

a) *k is an algebraic number field.*

Thm. 4 shows that $\text{vcd}(\Gamma_S) < \infty$. Since Γ is a subgroup of Γ_S , we have $\text{vcd}(\Gamma) \leq \text{vcd}(\Gamma_S) < \infty$, and since Γ is torsion-free, we have $\text{vcd}(\Gamma) = \text{cd}(\Gamma)$, whence the result.

b) *k is a function field over a finite field.*

We can no longer apply Thm. 4 directly. We let Γ act on the product T of the buildings T_v , $v \in S$. Since Γ is torsion-free, it acts *freely* on T , and we have $\text{cd}(\Gamma) \leq \dim(T)$, whence the result.

REMARK

In the preceding statement, it is essential that k be a *number field* (or a function field of one variable over a finite field). Thus, if we take for k the field $\mathbb{Q}(t)$, the subgroup Γ of $\text{GL}_2(k)$ generated by the matrices $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} t & 0 \\ 0 & 1 \end{pmatrix}$ is torsion-free and contains free abelian subgroups of arbitrarily large rank; we thus have $\text{cd}(\Gamma) = \infty$. I do not know if there exist analogous examples where Γ is *finitely presented*, and not just *finitely generated*.

2.5. Examples.

Let us begin with SL_2 :

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PROPOSITION 21. *Let k be an algebraic number field, and let S be a finite set of places of k containing the archimedean places. If Γ is an S -arithmetic subgroup of $\mathrm{SL}_2(k)$, we have*

$$\mathrm{vcd}(\Gamma) = n + s - 1, \quad \text{with } n = [k : \mathbf{Q}] \text{ and } s = \mathrm{Card}(S).$$

If $v \in S$, let us set $G_v = \mathrm{SL}_2(k_v)$ and let $G = \prod_{v \in S} G_v$. By Theorem 4, we have $\mathrm{vcd}(\Gamma) \leq d(G) - 1$, with $d(G) = \sum_{v \in S} d(G_v)$. If v is ultrametric, we have $d(G_v) = 1$, since SL_2 is of rank 1. If $v \in \Sigma^\infty$, and if $k_v \cong \mathbf{R}$, the symmetric space T_v attached to v is isomorphic to the hyperbolic plane $\mathrm{SL}_2(\mathbf{R})/\mathrm{SO}_2(\mathbf{R})$, and $d(G_v) = \dim(T_v) = 2$. If $k_v \cong \mathbf{C}$, we have

$$T_v \cong \mathrm{SL}_2(\mathbf{C})/\mathrm{SU}_2(\mathbf{C}) \quad \text{and} \quad d(G_v) = \dim(T_v) = 3.$$

We therefore have

$$d(G) = s + r_1 + 2r_2$$

where r_1 (resp. r_2) is the number of real (resp. complex) places of k ; since $n = r_1 + 2r_2$, we deduce

$$\mathrm{vcd}(\Gamma) \leq n + s - 1.$$

The reverse inequality results from the following lemma:

LEMMA 6. *Let Δ be an S -arithmetic subgroup of the upper triangular group of $\mathrm{SL}_2(k)$. We have $\mathrm{vcd}(\Delta) = n + s - 1$.*

[Since $\Delta \cap \Gamma$ is of finite index in Δ , we have

$$\mathrm{vcd}(\Delta) = \mathrm{vcd}(\Delta \cap \Gamma) \leq \mathrm{vcd}(\Gamma), \quad \text{cf. no. 1.8,}$$

and we indeed obtain the desired inequality.]

It remains to prove the lemma. Let ℓ be a prime number distinct from

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...residue characteristics of the local fields k_v , $v \in S$. According to the unit theorem, the group O_S^* of invertible elements of O_S is of rank $s - 1$. We can therefore find a subgroup A of O_S^* which is of finite index, and isomorphic to $\mathbb{Z}^{s-1}$; replacing A by a congruence subgroup if necessary, we can furthermore assume that all elements of A are congruent to 1 modulo ℓ . Let Δ' be the subgroup of $\mathrm{SL}_2(O_S)$ formed by the matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ such that $c = 0$ and $a \in A$. Since Δ' is commensurable with Δ , and torsion-free, we have $\mathrm{cd}(\Delta') = \mathrm{vcd}(\Delta)$ and we are reduced to showing that $\mathrm{cd}(\Delta') \geq n + s - 1$. Now, Δ' is an extension of A by O_S ; if we take $\mathbb{Z}/\ell\mathbb{Z}$ as coefficients, we thus have a spectral sequence:

$$E_2 = H^\bullet(A, H^\bullet(O_S, \mathbb{Z}/\ell\mathbb{Z})) \implies H^\bullet(\Delta', \mathbb{Z}/\ell\mathbb{Z}).$$

But we can write O_S as an extension of a group T by the group $\mathbb{Z}^n$, where T is a countable torsion group all of whose elements have order prime to ℓ . It follows easily that

$$H^\bullet(O_S, \mathbb{Z}/\ell\mathbb{Z}) = H^\bullet(\mathbb{Z}^n, \mathbb{Z}/\ell\mathbb{Z}),$$

and this cohomology algebra is therefore an exterior algebra with n independent generators of degree 1. Furthermore, the fact that the elements of A are congruent to 1 (mod ℓ) implies that A acts trivially on $H^\bullet(O_S, \mathbb{Z}/\ell\mathbb{Z})$. We can therefore write the term E_2 above in the form

$$E_2 = H^\bullet(A, \mathbb{Z}/\ell\mathbb{Z}) \otimes H^\bullet(O_S, \mathbb{Z}/\ell\mathbb{Z}),$$

and the two factors are exterior algebras having respectively $s - 1$ and n independent generators of degree 1. We deduce by a standard argument⁷ that $H^{n+s-1}(\Delta', \mathbb{Z}/\ell\mathbb{Z})$ is isomorphic to $\mathbb{Z}/\ell\mathbb{Z}$, whence $\text{cd}(\Delta') \geq n + s - 1$, which proves the lemma.

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REMARK

It would be interesting to find an *a priori* proof of the equality $\text{vcd}(\Gamma) = \text{vcd}(\Delta)$.

Let us now pass to *split groups* of rank ≥ 2 . Let k, S, n, s be as in Prop. 21, and let L be a split semisimple algebraic group of rank $\ell \geq 2$ and of dimension $N = \ell + 2m$. If $v \in S$, let $G_v = L(k_v)$. The invariant $d(G_v)$ is easy to calculate:

$$\begin{aligned} d(G_v) &= \ell && \text{if } v \text{ is ultrametric} \\ d(G_v) &= N - m = \ell + m && \text{if } v \text{ is real} \\ d(G_v) &= N = \ell + 2m && \text{if } v \text{ is complex.} \end{aligned}$$

If $G = \prod_{v \in S} G_v$, we therefore have

$$d(G) = \sum d(G_v) = \ell s + mr_1 + 2mr_2 = \ell s + mn.$$

Let Γ (resp. Δ) be an S -arithmetic subgroup of $L(k)$ (resp. of a Borel subgroup of $L(k)$). Theorem 4 shows that

$$\text{vcd}(\Gamma) < d(G) = \ell s + mn.$$

On the other hand, an argument analogous to that of Lemma 6 gives:

$$\text{vcd}(\Delta) = \ell(s - 1) + mn = \ell s + mn - \ell.$$

We conclude as above:

$$\ell s + mn - \ell \leq \text{vcd}(\Gamma) < \ell s + mn. \quad (*)$$

Since $\ell \geq 2$, the inequalities $(*)$ are not sufficient to determine the value of $\text{vcd}(\Gamma)$. Thus, for $\text{SL}_3(\mathbf{Z})$, we have $\ell = 2$, $m = 3$, $n = 1$, $s = 1$, whence:

$$\text{vcd}(\text{SL}_3(\mathbf{Z})) = 3 \quad \text{or} \quad 4.$$

Similarly:

$$\text{vcd}(\text{Sp}_4(\mathbf{Z})) = 4 \quad \text{or} \quad 5.$$

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In fact, it is the lowest values that are correct. More generally, one can show that, if G, L, Γ satisfy the hypotheses of Th. 4, we have $\text{vcd}(\Gamma) = d(G) - \ell$, where ℓ is the rank of L over k (the proof will appear in a work written in collaboration with A. Borel).

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⁷We can even show that $E_2 = E_\infty$ and that $H^\bullet(\Delta', \mathbb{Z}/\ell\mathbb{Z})$ is an exterior algebra with $n + s - 1$ independent generators of degree 1.

§3. EULER-POINCARÉ MEASURES

3.1. Definition.

Let G be a locally compact unimodular group (Bourbaki, *Int.*, chap. VII, p. 18) and let μ be an invariant measure on G . If Γ is a discrete subgroup of G , the measure μ defines by passing to the quotient a measure on G/Γ , also denoted by μ ; if this measure is bounded, we can therefore define

$$\mu(G/\Gamma) = \int_{G/\Gamma} \mu;$$

this applies in particular when G/Γ is compact.

DEFINITION. We say that μ is an **Euler-Poincaré measure** on G if every torsion-free discrete subgroup Γ of G with compact quotient has the following two properties:

- (a) Γ is of type (FL).
- (b) We have $\chi(\Gamma) = \mu(G/\Gamma)$.

REMARKS.

- 1) Suppose that μ is an Euler-Poincaré measure on G , and let Γ be a discrete subgroup of G with compact quotient. Suppose that Γ is virtually torsion-free (which is the case for all “usual” groups G). Then Γ is of type (WFL) and we have $\chi(\Gamma) = \mu(G/\Gamma)$; this is seen by applying (a) and (b) above to a torsion-free subgroup of finite index in Γ .
- 2) When G has a torsion-free discrete subgroup with compact quotient, there exists *at most one* Euler-Poincaré measure on G . If there is one, we call it *the* Euler-Poincaré measure of G .
- 3) We have seen in §2 many examples of groups G for which every torsion-free discrete subgroup with compact quotient is of type (FL); for such a group, the only property to verify is (b).

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Let us now give some examples of Euler-Poincaré measures (others will be seen subsequently):

- (i) A discrete group G of type (WFL) has an Euler-Poincaré measure μ characterized by the formula

$$\mu(\{g\}) = \chi(G) \quad \text{for all } g \in G.$$

- (ii) If G is compact, the Euler-Poincaré measure of G is the Haar measure of G normalized by the condition $\mu(G) = 1$.

- (iii) The Euler-Poincaré measure of $\mathbf{R}^n$ is zero if $n \geq 1$.

- (iv) Let k be a local field, with finite residue field having q elements, and let O_k be the ring of integers of k . Let $G = \mathrm{SL}_2(k)$, and let μ be the Haar measure of G , normalized in such a way that the open compact subgroup $\mathrm{SL}_2(O_k)$ has measure $q - 1$, cf. [38], p. II-20. The Euler-Poincaré measure of G is equal to $-\mu$. Indeed, according to Ihara [22], G contains torsion-free discrete subgroups with compact quotient, and if Γ is such a subgroup, Γ is a free group of rank $\mu(G/\Gamma) - 1$ ([22], Th. 1 — see also [38], loc. cit., Th. 5); we thus have

$$\chi(\Gamma) = 1 - \mathrm{rg}(\Gamma) = -\mu(G/\Gamma)$$

which indeed shows that $-\mu$ is the Euler-Poincaré measure of G .

3.2. The real case (reminders).

This is the case where G is a real, unimodular Lie group having a finite number of connected components. If K is a maximal compact subgroup of G , we denote by T the homogeneous space G/K , cf. no. 2.1. It is well known (cf. for example [32], [34]) that the application of the Gauss-Bonnet formula to the quotients of T implies the existence of an Euler-Poincaré measure on G . More precisely, let us distinguish two cases:

a) $d(G) = \dim(T)$ is odd.

In this case, if Γ is a torsion-free discrete subgroup of G with compact quotient, the manifold $X_\Gamma = \Gamma \backslash T = \Gamma \backslash G/K$ is a compact manifold of odd dimension, and its Euler-Poincaré characteristic is

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is zero. We thus have $\chi(\Gamma) = 0$, which shows that 0 is an Euler-Poincaré measure on G .

b) $d(G) = \dim(T)$ is even.

Let $d(G) = 2n$. Equip T with a Riemannian structure invariant under G , which is possible since K is compact. With this structure is associated an Euler-Poincaré form Ω which is a twisted form of degree $2n$ on T , and thus identifies with a real measure on T . Let us briefly recall how this form is defined:

In general, let X be a Riemannian manifold of dimension $2n$ and let R be its curvature tensor (cf. Milnor [30] §9). Let $e_1, \dots, e_{2n}$ be an orthonormal frame of the tangent bundle of X in an open set U . On U , the data of R is equivalent to that of a family (Ω_{ij}) , $1 \leq i, j \leq 2n$, of differential forms of degree 2, such that

$$R(u, v)e_i = \sum_j \Omega_{ij}(u, v)e_j$$

for every pair (u, v) of vector fields on U . The forms Ω_{ij} depend in an alternating fashion on (i, j) and commute with each other, since they are of degree 2. We can therefore define the pfaffian $\text{Pf}((\Omega_{ij}))$ of the matrix (Ω_{ij}) , cf. Bourbaki, *Alg.*, chap. IX, §5, $n^\circ 2$. This is a form of degree $2n$ on U . Let us set

$$\Omega = \frac{1}{(2\pi)^n} \text{Pf}((\Omega_{ij})).$$

If we replace $e_1, \dots, e_{2n}$ by another orthonormal frame of the same orientation (resp. of opposite orientation), the form Ω does not change (resp. is replaced by $-\Omega$). It therefore defines by gluing a twisted form of degree $2n$ on X , i.e. a real measure, called the Gauss-Bonnet form (or measure). If A is a compact piece of X (Bourbaki, *Var.*, R., 11.1), with boundary ∂A , the Gauss-Bonnet formula takes the following form (cf. Chern [14]):

$$\chi(A) = \int_A \Omega + \int_{\partial A} \Pi,$$

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where Π is a certain form of degree $2n - 1$ on ∂A . In particular, if X is compact, we can take $A = X$, $\partial A = \emptyset$, and we obtain the usual Gauss-Bonnet formula:

$$\chi(X) = \int_X \Omega.$$

Let us return now to the case of the manifold $T = G/K$, endowed with a Riemannian structure invariant under G . The corresponding form Ω is also invariant under G . Since K is compact, there exists a unique invariant measure μ on G whose image under $G \rightarrow G/K$ is the measure Ω ; if ν denotes the Haar measure of K , normalized so that $\nu(K) = 1$, we have $\mu/\nu = \Omega$.

Proposition (22). *The measure μ is an Euler-Poincaré measure on G .*

Indeed, let Γ be a torsion-free discrete subgroup of G with compact quotient. If Ω_Γ denotes the Gauss-Bonnet form of the Riemannian space $X_\Gamma = \Gamma \backslash T$, we have $\chi(\Gamma) = \chi(X_\Gamma) = \int_{X_\Gamma} \Omega_\Gamma$. But the

local nature of Ω shows that the pullback of Ω_Γ under $T \rightarrow X_\Gamma = \Gamma \backslash T$ is Ω . We conclude that the image under the projection $\Gamma \backslash G \rightarrow X_\Gamma$ of the measure μ is Ω_Γ , whence

$$\chi(\Gamma) = \int_{X_\Gamma} \Omega_\Gamma = \int_{\Gamma \backslash G} \mu = \mu(\Gamma \backslash G).$$

The proposition follows from this, and from the fact that $\mu(\Gamma \backslash G) = \mu(G/\Gamma)$ since G is unimodular.

Let us pass now to the case where G is reductive, i.e., of the form $L(\mathbf{R})$ where L is a reductive algebraic group over $\mathbf{R}$. Let $\mathfrak{g}$ (resp. $\mathfrak{k}$) denote the Lie algebra of G (resp. of K). The following result is well known:

Proposition (23). *The Euler-Poincaré measure μ of G is $\neq 0$ if and only if $\mathfrak{g}$ and $\mathfrak{k}$ have the same rank (i.e., if $\mathfrak{k}$ contains a subalgebra*

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of Cartan of $\mathfrak{g}$). If this is so, the sign of μ is $(-1)^n$, where

$$n = \frac{1}{2} \dim(G/K).$$

(Note that G possesses discrete subgroups with compact quotient, according to a theorem of Borel [5]; we can therefore speak of the Euler-Poincaré measure of G .)

The proof rests on the proportionality principle of Hirzebruch. Let us summarize it:

We first easily reduce to the case where G is semi-simple. Let $\mathfrak{m}$ be the orthogonal complement of $\mathfrak{k}$ in $\mathfrak{g}$ with respect to the Killing form. If we identify $\mathfrak{m}$ with the tangent space to G/K at the identity element, the curvature tensor R at this point is given by

$$R(u, v)w = [[u, v], w], \quad (u, v, w \in \mathfrak{m})$$

cf. for example [25], p. 231, th. 3.2 (note that the definition of R used in [25] is the opposite of that of Milnor [30], which we have adopted). Let, on the other hand, $\mathfrak{g}_\mathbf{C} = \mathbf{C} \otimes \mathfrak{g} = \mathfrak{g} \oplus i\mathfrak{g}$ be the complexification of $\mathfrak{g}$. The decomposition of $\mathfrak{g}$ into $\mathfrak{k} \oplus \mathfrak{m}$ gives a decomposition of $\mathfrak{g}_\mathbf{C}$:

$$\mathfrak{g}_\mathbf{C} = \mathfrak{k} \oplus \mathfrak{m} \oplus i\mathfrak{k} \oplus i\mathfrak{m}.$$

The Lie algebra $\mathfrak{g}' = \mathfrak{k} \oplus i\mathfrak{m}$ is a real form of $\mathfrak{g}_\mathbf{C}$. Let G' be the corresponding simply connected group, and let K' be the subgroup of G' corresponding to the subalgebra $\mathfrak{k}$. The groups G' and K' are compact; the homogeneous space G'/K' is sometimes called the dual of the space G/K . If we identify $i\mathfrak{m}$ with the tangent space to G'/K' at the identity element, the curvature tensor R' of G'/K' is given by the same formula as above:

$$R'(u, v)w = [[u, v], w], \quad (u, v, w \in i\mathfrak{m}).$$

It follows that the bijection $u \mapsto iu$ transforms R into $-R'$. Let us denote

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then Ω (resp. Ω') is the value at the identity element of the Gauss-Bonnet form of G/K (resp. of G'/K'). The preceding shows that $u \mapsto iu$ transforms Ω into $(-1)^n \Omega'$, with $n = \frac{1}{2} \dim(G/K)$ if $\dim(G/K)$ is even (if this dimension is odd, we have $\Omega = \Omega' = 0$). But the integral of Ω' over G'/K' is equal to $\chi(G'/K')$. Prop. 24 thus amounts to asserting that $\chi(G'/K') = 0$ if the rank of $\mathfrak{k}$ is strictly less than that of $\mathfrak{g}$, and that $\chi(G'/K') > 0$ if these two ranks are equal, which is a classical result of Hopf-Samelson [21].

Examples

- 1) Suppose that G is the real group underlying a complex reductive group of dimension ≥ 1 . The rank of $\mathfrak{k}$ is then equal to half of that of $\mathfrak{g}$; in view of the preceding proposition, we conclude that $\mu = 0$.

- 2) If G is split semi-simple, we have $\mu \neq 0$ if and only if -1 belongs to the Weyl group of the root system of G ; this is equivalent to saying that all simple factors of $\mathfrak{g}$ are of type A_1 , B_ℓ , C_ℓ , D_ℓ (ℓ even), E_7 , E_8 , F_4 or G_2 .
- 3) Let us take in particular $G = \mathrm{SL}_2(\mathbf{R})$. Let

$$X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

be the canonical basis of $\mathfrak{g}$ (and even of $\mathfrak{g}_{\mathbf{Z}}$), and let a be the 3-covector on $\mathfrak{g}$ such that $a(X, Y, Z) = 1$. This covector defines on G a Haar measure μ_a ("arithmetic" measure). In terms of μ_a , the Euler-Poincaré measure μ of G is given by the formula

$$\mu = -\frac{1}{2\pi^2} \mu_a.$$

This is verified, for example, by identifying the homogeneous space

$$T = \mathrm{SL}_2(\mathbf{R})/\mathrm{SO}_2(\mathbf{R})$$

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to the Poincaré half-plane $H = \{z \mid \mathrm{Im}(z) > 0\}$, and by showing that the Gauss-Bonnet measure of H is $\frac{-1}{2\pi} dx dy / y^2$.

3.3. The ultrametric case

In this section, G denotes a locally compact unimodular group which acts cellularly on a CW-complex T . We make the following hypotheses:

- i) T is contractible.
- ii) T is locally compact (hence finite-dimensional).
- iii) For every cell σ of T , the stabilizer G_σ of σ in G is an open compact subgroup of G .
- iv) Every compact subgroup of G is contained in some G_σ .
- v) The cells of T are finite in number modulo the action of G .

(These conditions are satisfied when $G = \mathbf{L}(k)$, where k is a locally compact local field and $\mathbf{L}$ is a connected reductive algebraic group defined over k , the complex T being the building of G in the sense of section 2.2.)

Let $\mathfrak{S}$ be the set of cells of T and let Σ be a system of representatives of $G \backslash \mathfrak{S}$ in $\mathfrak{S}$; by v), Σ is finite. If μ is a non-zero invariant measure on G , we set

$$\chi(\mu) = \sum_{\sigma \in \Sigma} (-1)^{\dim(\sigma)} \frac{1}{\mu(G_\sigma)}.$$

This sum makes sense, since each of the G_σ is open and compact, hence has non-zero measure. It does not depend on the chosen system of representatives Σ : indeed, since G is unimodular, we have

$$\mu(G_{g\sigma}) = \mu(gG_\sigma g^{-1}) = \mu(G_\sigma)$$

if $g \in G$, $\sigma \in \mathfrak{S}$. Moreover, if we multiply μ by a constant c , $\chi(\mu)$ is multiplied by c^{-1} . The product $\mu_G = \chi(\mu)\mu$ is therefore independent of the choice of μ ; we shall call it the *canonical measure* of the group G .

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PROPOSITION 24. *The canonical measure μ_G is an Euler-Poincaré measure.*

Let Γ be a torsion-free discrete subgroup of G , with compact quotient. The group Γ acts freely and properly on T and the quotient $\Gamma \backslash T$ is compact. By Prop. 9 (or Prop. 11), Γ is of type (FL), and we have

$$\chi(\Gamma) = \chi(\Gamma \backslash T) = \sum_{\sigma \in \Gamma \backslash \mathfrak{S}} (-1)^{\dim(\sigma)}.$$

But the G -set $\mathfrak{S}$ identifies with the disjoint union of the homogeneous spaces G/G_σ , $\sigma \in \Sigma$, and $\Gamma \backslash \mathfrak{S}$ identifies with the disjoint union of the $\Gamma \backslash G/G_\sigma$, $\sigma \in \Sigma$. We can therefore rewrite the expression above in the form

$$\chi(\Gamma) = \sum_{\sigma \in \Sigma} (-1)^{\dim(\sigma)} \text{Card}(\Gamma \backslash G/G_\sigma).$$

Since Γ is torsion-free, the open compact group G_σ acts freely (on the right) on $\Gamma \backslash G$. We conclude that, if μ is a Haar measure of G , we have

$$\mu(\Gamma \backslash G) = \mu(G_\sigma) \cdot \text{Card}(\Gamma \backslash G/G_\sigma).$$

From which:

$$\begin{aligned} \chi(\Gamma) &= \mu(\Gamma \backslash G) \sum_{\sigma \in \Sigma} (-1)^{\dim(\sigma)} \frac{1}{\mu(G_\sigma)} \\ &= \mu(\Gamma \backslash G) \chi(\mu) = \mu_G(G/\Gamma), \end{aligned}$$

which proves the proposition.

COROLLARY. *Let L be a connected reductive group over a locally compact local field K . The group $G = L(k)$ has an Euler-Poincaré measure.*

We shall see in the following section that, if L is semisimple, this measure is $\neq 0$ and its sign is $(-1)^\ell$, where ℓ is the rank of L over k (i.e., the dimension of T).

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EXAMPLE

Let us take $G = \text{SL}_2(k)$, so that T is a tree and Σ consists of two vertices P, Q and the edge PQ . If q is the number of elements of the residue field of k , the group G_{PQ} is of index $q+1$ in G_P and G_Q . Whence:

$$\begin{aligned} \chi(\mu) &= 1/\mu(G_P) + 1/\mu(G_Q) - 1/\mu(G_{PQ}) \\ &= (1 + 1 - q - 1)/\mu(G_P) \\ &= (1 - q)/\mu(G_P). \end{aligned}$$

The Euler-Poincaré measure μ_G is therefore characterized by the relation $\mu_G(G_P) = -(q-1)$; since $G_P = \text{SL}_2(O_k)$, we recover the result of example (iv) of no. 3.1.

REMARK

It is very likely that every group G of the form $L(k)$ has torsion-free discrete subgroups with compact quotient; this should be provable by arithmetic means, as Borel [5] did in the real case.

Euler-Poincaré measure on a product $H \times G$

Let H be a locally compact unimodular group, equipped with an invariant measure μ_H . Let us equip the group $H \times G$ with the product measure $\mu_{H \times G} = \mu_H \otimes \mu_G$ of μ_H by the canonical measure of G . Let Γ be a torsion-free discrete subgroup of $H \times G$. For any $\sigma \in \mathfrak{S}$, let

$$\Gamma_\sigma = \Gamma \cap (H \times G_\sigma);$$

the projection $\Gamma_\sigma \rightarrow H$ is injective (since its kernel is discrete and compact); it identifies Γ_σ with a torsion-free discrete subgroup of H . Let us make the following hypotheses:

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- (a) For all $\sigma \in \mathfrak{S}$, the set $\Gamma \backslash (H \times G)/(H \times G_\sigma)$ is finite.
(b) For all $\sigma \in \mathfrak{S}$, the group Γ_σ is of type (FL), the quotient H/Γ_σ has finite volume, and $\chi(\Gamma_\sigma) = \mu_H(H/\Gamma_\sigma)$.

PROPOSITION 25. *Under the assumptions above, Γ is of type (FL), the quotient $(H \times G)/\Gamma$ has finite volume, and $\chi(\Gamma) = \mu_{H \times G}((H \times G)/\Gamma)$.*

The proof is analogous to that of Prop. 24. The fact that Γ is of type (FL) follows from Prop. 20, or from Prop. 11. The latter further shows that

$$\chi(\Gamma) = \sum_{\sigma \in \Gamma \backslash \mathfrak{S}} (-1)^{\dim(\sigma)} \chi(\Gamma_\sigma),$$

whence

$$\chi(\Gamma) = \sum_{\sigma \in \Gamma \backslash \mathfrak{S}} (-1)^{\dim(\sigma)} \mu_H(\Gamma_\sigma \backslash H).$$

Let I_σ be a system of representatives for the elements of

$$\Gamma \backslash (H \times G)/(H \times G_\sigma).$$

The $g\sigma$ ($\sigma \in \Sigma$, $g \in I_\sigma$) form a system of representatives for $\Gamma \backslash \mathfrak{S}$. Thus we have:

$$\chi(\Gamma) = \sum_{\sigma \in \Sigma} (-1)^{\dim(\sigma)} \sum_{g \in I_\sigma} \mu_H(\Gamma_{g\sigma} \backslash H).$$

Let us set:

$$\Gamma_\sigma^g = g^{-1} \Gamma_{g\sigma} g = (H \times G_\sigma) \cap g^{-1} \Gamma g.$$

Let $H \times G_\sigma$ act on the right on $\Gamma \backslash (H \times G)$; this space decomposes into orbits isomorphic to $\Gamma_\sigma^g \backslash (H \times G_\sigma)$, $g \in I_\sigma$. We deduce that $(H \times G)/\Gamma$ has finite volume, and that, if μ is a Haar measure on G , we have

$$(\mu_H \otimes \mu)(\Gamma \backslash (H \times G)) = \sum_{g \in I_\sigma} (\mu_H \otimes \mu)(\Gamma_\sigma^g \backslash (H \times G_\sigma)).$$

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Furthermore, by projecting $\Gamma_\sigma^g \backslash (H \times G_\sigma)$ onto $\Gamma_\sigma^g \backslash H$, we obtain:

$$\begin{aligned} (\mu_H \otimes \mu)(\Gamma_\sigma^g \backslash (H \times G_\sigma)) &= \mu_H(\Gamma_\sigma^g \backslash H) \cdot \mu(G_\sigma) \\ &= \mu_H(\Gamma_{g\sigma} \backslash H) \cdot \mu(G_\sigma). \end{aligned}$$

Using these formulas, we see that we can rewrite $\chi(\Gamma)$ in the form

$$\begin{aligned} \chi(\Gamma) &= (\mu_H \otimes \mu)(\Gamma \backslash (H \times G)) \cdot \sum_{\sigma \in \Sigma} (-1)^{\dim(\sigma)} \frac{1}{\mu(G_\sigma)} \\ &= (\mu_H \otimes \mu)(\Gamma \backslash (H \times G)) \cdot \chi(\mu) \\ &= (\mu_H \otimes \mu_G)((H \times G)/\Gamma), \end{aligned}$$

which proves the proposition.

COROLLARY. *If μ_H is an Euler-Poincaré measure on H , the measure $\mu_{H \times G} = \mu_H \otimes \mu_G$ is an Euler-Poincaré measure on $H \times G$.*

Let Γ be a torsion-free discrete subgroup of $H \times G$ with compact quotient. If $\sigma \in \mathfrak{S}$, the space $\Gamma \backslash (H \times G)/(H \times G_\sigma)$ is compact and discrete, hence finite. On the other hand, we see, as in the proof of Th. 3 of No. 2.3, that H/Γ_σ^g is compact. Given the hypothesis made on (H, μ_H) , this implies that Γ_σ^g is of type (FL) and that $\chi(\Gamma_\sigma^g)$ is equal to $\mu_H(H/\Gamma_\sigma^g)$. We can therefore apply Prop. 25 to Γ ; hence the fact that $\mu_{H \times G}$ is an Euler-Poincaré measure on $H \times G$.

3.4. The ultrametric case: determination of μ_G

Let k be a locally compact local field, let L be a simply connected simple group over k , of relative rank ℓ , and let G be the group $L(k)$. We propose to determine the canonical measure μ_G of G . Since μ_G is invariant, it suffices to know its value on a subgroup

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compact open subset of G . Now, according to the results of Bruhat-Tits recalled in no. 2.2, the group G possesses a *Tits system* (G, B, N, S) (cf. [9], p. 22) in which B is a compact open subgroup of G , called the *Iwahori subgroup*. It therefore suffices to determine $\mu_G(B)$. This is done by means of a certain rational function of several variables that we shall define (the same function appears in the work of I. G. Macdonald on the Plancherel formula):

The function $W(t)$

Let (W, S) be a Coxeter system ([9], p. 11) such that S is *finite*. Let $t = (t_i)_{i \in I}$ be a finite family of indeterminates, and let us be given a map $s \mapsto i(s)$ from S to I satisfying the following equivalent conditions (cf. [9], p. 12, prop. 3):

- (α) $i(s) = i(s')$ if s and s' are conjugate in W ;
- (β) $i(s) = i(s')$ if the order $m(s, s')$ of ss' is finite and odd.

If $s \in S$, we write t_s instead of $t_{i(s)}$.

Let $w \in W$. Let us choose a reduced decomposition $(s_1, \dots, s_q)$ of w , cf. [9], p. 9; *the monomial $t_w = t_{s_1} \dots t_{s_q}$ is independent of the choice of $(s_1, \dots, s_q)$* by virtue of Prop. 5, p. 16 of [9]. The total degree of t_w is equal to the length $\ell(w)$ of w ; it tends to infinity with w . If A is a subset of W , we can therefore set:

$$A(t) = A((t_i)_{i \in I}) = \sum_{w \in A} t_w.$$

This is a formal power series with integer coefficients ≥ 0 in the t_i ; when the family (t_i) is reduced to a single element t , we recover the series $A(t)$ studied in [9], p. 45, exerc. 26.

If Y is a subset of S and W_Y the subgroup it generates, we can apply the preceding to $A = W_Y$. We thus obtain a formal power series $W_Y(t)$; if $Y = S$, we write $W(t)$ instead of $W_S(t)$.

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PROPOSITION 26. (a) $W(\mathbf{t})$ is a rational function of the t_i .

(b) If W is finite, and if w_0 is its element of greatest length, we have

$$t_{w_0}/W(\mathbf{t}) = 1/W(\mathbf{t}^{-1}) = \sum_{Y \subset S} \varepsilon(Y)/W_Y(\mathbf{t}), \quad \text{where } \varepsilon(Y) = (-1)^{\text{Card}(Y)}.$$

(c) If W is infinite, we have

$$0 = \sum_{Y \subset S} \varepsilon(Y)/W_Y(\mathbf{t}).$$

(d) If W is infinite, and if all the W_Y ($Y \neq S$) are finite, we have

$$W(\mathbf{t}^{-1}) = -\varepsilon(S)W(\mathbf{t}).$$

(e) If W is of Euclidean type, the series $W(\mathbf{t})$ converges absolutely in the unit polydisc $|t_i| < 1$.

The proofs of (b) and (c) are identical to those of the corresponding results in Exercise 26, *loc. cit.* (see also Solomon [42], §3). It is unnecessary to reproduce them. Assertion (a) is proved by induction on $\text{Card}(S)$. The case where W is finite is trivial, $W(\mathbf{t})$ being a polynomial. If W is infinite, (c) shows that

$$\varepsilon(S)/W(\mathbf{t}) = - \sum_{Y \neq S} \varepsilon(Y)/W_Y(\mathbf{t})$$

and the $W_Y(\mathbf{t})$ are rational functions by virtue of the induction hypothesis; the same is therefore true for $W(\mathbf{t})$.

For (d), we substitute $\mathbf{t}^{-1}$ for $\mathbf{t}$ in (c); this gives the formula

$$\varepsilon(S)/W(\mathbf{t}^{-1}) = - \sum_{Y \neq S} \varepsilon(Y)/W_Y(\mathbf{t}^{-1}).$$

Combining this with (b), applied to the W_Y , we obtain:

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$$\varepsilon(S)/W(t^{-1}) = - \sum_{Y \neq S} \varepsilon(Y) \sum_{Z \subset Y} \varepsilon(Z)/W_Z(t).$$

But, if Z is a subset of S distinct from S , the sum of the $\varepsilon(Y)$, for $Z \subset Y \subset S$, is zero; the analogous sum, extended to Y distinct from S , is therefore equal to $-\varepsilon(S)$. This allows us to rewrite the preceding formula in the form

$$\varepsilon(S)/W(t^{-1}) = -\varepsilon(S) \sum_{Z \neq S} \varepsilon(Z)/W_Z(t),$$

and by comparing with (c), we indeed deduce that $1/W(t^{-1})$ is equal to $-\varepsilon(S)/W(t)$.

It remains to prove (e). Since the coefficients of the series $W(t)$ are ≥ 0 , it suffices to show that the series in one variable $W(t)$ converges absolutely in the disk $|t| < 1$; this follows from the fact that the poles of $W(t)$ are roots of unity, cf. for example [23], no. 1.8 or [9], p. 231.

Application to Tits systems

Let (G, B, N, S) be a Tits system with Weyl group $W = N/(B \cap N)$. If $w \in W$, let us denote by $C(w)$ the double coset BwB ; the group G is the disjoint union of the $C(w)$, $w \in W$ (Bruhat decomposition). If $Y \subset S$, let us denote by G_Y the union of the $C(w)$, $w \in W_Y$; this is the parabolic subgroup of G defined by Y .

Suppose that all $C(w)$ are finite unions of left cosets modulo B , and let us set

$$q_w = \text{Card}(C(w)/B) = (B : B_w) \quad \text{where} \quad B_w = B \cap wBw^{-1}.$$

This applies in particular to the elements s of S .

Lemma (7). (a) *The q_s are integers ≥ 2 .*

(b) *If $(s_1, \dots, s_q)$ is a reduced decomposition of w , we have*

$$q_w = q_{s_1} \cdots q_{s_q}.$$

(c) *If $s, s' \in S$ are conjugate in W , we have $q_s = q_{s'}$.*

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This result is known, cf. [9], p. 55, exerc. 24. Let us recall its proof:

If q_s were equal to 1, we would have $BsB = sB$ and, by passing to inverses, $BsB = Bs$, whence $sBs = ssB = B$, contrary to axiom (T_4) of Tits systems ([9], p. 22). This proves (a).

For (b), it suffices to prove that, if $w \in W$ and $s \in S$ are such that $\ell(ws) = \ell(w) + 1$, we have $q_{ws} = q_w q_s$. Now, in this case, we know that $C(ws) = C(w)C(s)$, cf. [9], p. 26, cor. 1. Let us then decompose $C(w)$ (resp. $C(s)$) into a disjoint union of left cosets $g_i B$ (resp. $h_j B$), with $1 \leq i \leq q_w$ (resp. $1 \leq j \leq q_s$). It follows from the formula above that every element of $C(ws)$ is contained in a left coset $g_i h_j B$ and it all comes down to showing that these cosets are pairwise distinct. Suppose then that we have $g_i h_j \in g_{i'} h_{j'} B$. Since h_j and $h_{j'}$ belong to the group $G_s = B \cup C(s)$, we have $g_i \in g_{i'} G_s$ whence $g_i \in g_{i'} B$ or $g_i \in g_{i'} C(s)$. The second relation is impossible, because it would imply

$g_i \in C(w)C(s) = C(ws)$ whereas g_i belongs to $C(w)$. We thus have $g_i \in g_{i'}B$, i.e., $g_i = g_{i'}$, whence $h_jB = h_{j'}B$ and $j = j'$, which proves our assertion [when we no longer make finiteness assumptions, the same argument shows that $C(ws)$ is identified with $C(w) \times^B C(s)$ if $\ell(ws) = \ell(w) + 1$].

To prove (c), it suffices to see that, if the order n of ss' is finite and odd, we have $q_s = q_{s'}$. Let $n = 2m + 1$, and let $w = (ss')^m s = (s's)^m s'$; the decompositions

$$(s, s', s, s', \dots, s) \quad \text{and} \quad (s', s, s', s, \dots, s')$$

of w are both reduced ([9], p. 11). Applying (b) to them, we see that

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$$(q_s q_{s'})^{m_{ss'}} = q_w = (q_{s'} q_s)^{m_{s's}}$$

hence $q_s = q_{s'}$, which completes the proof of the lemma.

Now let I be the quotient of S by the relation

$$“s \text{ and } s' \text{ are conjugate in } W,”$$

let i be the canonical map from S to I , and let $t = (t_i)_{i \in I}$ be a family of indeterminates indexed by I . If $s \in S$, set, as above, $t_s = t_{i(s)}$ and $W(t) = \sum_{w \in W} t_w$. By Lemma 7, we can define a family $q = (q_i)_{i \in I}$ of integers ≥ 2 such that $q_{i(s)} = q_s = \text{Card}(C(s)/B)$ for all $s \in S$; the value $t_w(q)$ of the monomial t_w for the family (q) is equal to q_w , by (b).

Proposition (27). *Suppose W is infinite, of irreducible Euclidean type.*

(a) *The rational function $W(t)$ is defined at the points q and q^{-1} ; we have*

$$-\varepsilon(S)W(q) = W(q^{-1}) = \sum_{w \in W} 1/q_w,$$

this latter series being convergent.

(b) *If Y is a subset of S distinct from S , the index of B in G_Y is finite, and equal to $W_Y(q) = \sum_{w \in W_Y} q_w$.*

(c) *We have $\sum_{Y \neq S} \varepsilon(Y)/(G_Y : B) = -\varepsilon(S)/W(q) > 0$.*

Since the q_s are ≥ 2 , Prop. 26 (e) shows that $W(t)$ is defined at the point q^{-1} and that its value at this point is equal to the sum of the convergent series $\sum 1/q_w$; on the other hand, the W_Y are finite if $Y \neq S$, and Prop. 26 (d) shows that

$$W(t^{-1}) = -\varepsilon(S)W(t),$$

whence (a).

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Assertion (b) follows from the fact that G_Y is the disjoint union of the $C(w)$, $w \in W_Y$, whence $\text{Card}(G_Y/B) = \sum_{w \in W_Y} \text{Card}(C(w)/B) = W_Y(q)$.

Finally, we obtain (c) by setting $t = q$ in the formula

$$-\varepsilon(S)/W(t) = \sum_{Y \neq S} \varepsilon(Y)/W_Y(t),$$

cf. Prop. 26 (c); the fact that $-\varepsilon(S)/W(q) = 1/W(q^{-1})$ is > 0 follows from the expression of $W(q^{-1})$ as a series given in (a) [note that the same series appears in [29]].

The case of a simply connected simple group over a local field

Let us return to the group $G = L(k)$ introduced at the beginning of this section. One can apply to the Tits system (G, B, N, S) all the properties proved above. We have $\text{Card}(S) = \ell + 1$, where ℓ is the rank of L . The integers q_s are powers of the number q of the residue field of k ; the number $W(q)$ is defined and non-zero; its sign is $(-1)^\ell$.

Theorem (6). *We have $\mu_G(B) = 1/W(q) = (-1)^\ell/W(q^{-1})$.*

(Recall that μ_G denotes the canonical measure on G .)

Let μ be a Haar measure on G ; we need to compute $\chi(\mu)$, cf. no. 3.3. However, by the very definition of the building T , it contains a simplex C of dimension ℓ , whose vertices (x_s) are indexed by S . If $Y \subset S$, $Y \neq S$, let us denote by C_Y the face of C with vertices x_s , $s \in S - Y$; the stabilizer of C_Y is G_Y ; moreover, the set of C_Y , $Y \neq S$, constitutes a system of representatives for the cells of T modulo G . Thus we have

$$\chi(\mu) = \sum_{Y \neq S} -\varepsilon(S - Y)/\mu(G_Y),$$

or indeed, since $\mu(G_Y) = \mu(B) \cdot (G_Y : B)$,

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$$\chi(\mu) = \frac{-\varepsilon(S)}{\mu(B)} \sum_{Y \neq S} \varepsilon(Y)/(G_Y : B).$$

Applying Prop. 27 (c) we obtain

$$\chi(\mu) = 1/W(q)\mu(B),$$

i.e.

$$1/W(q) = \chi(\mu)\mu(B) = \mu_G(B),$$

hence the theorem, by Prop. 27 (a).

COROLLARY. *The measure μ_G is $\neq 0$; its sign is $(-1)^\ell$.*

Indeed, we have seen that $W(q^{-1}) = \sum 1/q_w$ is > 0 .

Split case

This is the case treated by Iwahori-Matsumoto [23]. We start with a group scheme L_0 over O_k , which is assumed to be simple, simply connected and split; let R be its root system and W_R its Weyl group. We can apply Thm. 6 to the algebraic group $L = L_0 \times_{O_k} k$ obtained from L_0 by extension of scalars. The group W is then the affine Weyl group $\tilde{W}_R$ of W_R and S is the set of vertices of the completed Dynkin diagram ([9], p. 198) of R ; in particular, there is an element s_0 of S such that $Y = S - \{s_0\}$ is the Coxeter graph of R ; the corresponding group G_Y is equal to $L_0(O_k)$; it is a maximal compact subgroup of $G = L(k)$.

THEOREM 7. *Let $m_1, \dots, m_\ell$ be the exponents ([9], p. 118) of W_R and let q be the number of elements of the residue field of k . We have*

$$\mu_G(B) = \prod_{i=1}^{i=\ell} \frac{1 - q}{1 + q + \dots + q^{m_i}}$$

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and

$$\mu_G(L_0(O_k)) = \prod_{i=1}^{i=\ell} (1 - q^{-m_i}).$$

The integers q_s are all equal to q here, cf. [23], 3.1. To compute $\mu_G(B) = 1/W(q)$, we can therefore replace the function of several variables $W(t)$ by the function $W(t) = \sum_{w \in W} t^{\ell(w)}$.

As we have

$$W(q) = \prod_{i=1}^{i=\ell} (1 + q + \dots + q^{m_i})/(1 - q^{-m_i}),$$

cf. [23], 1.10 and [9], p. 231, we indeed find the announced value for $\mu_G(B)$.

On the other hand, the index of B in $L_0(O_k) = G_Y$ is equal to $W_Y(q)$, cf. Prop. 27 (b). We therefore have

$$\mu_G(L_0(O_k)) = W_Y(q)\mu_G(B).$$

But W_Y is equal to W_R , and we have

$$W_R(q) = \prod_{i=1}^{i=\ell} (1 + q + \cdots + q^{m_i}),$$

cf. [23], [9], *loc. cit.* Whence the desired result.

Reductive groups

We now pass to the case of a *reductive* group L over k (not necessarily simple), and let $G = L(k)$.

Proposition (28). *The canonical measure of G is $\neq 0$ if and only if the rank over k of the center of L is zero (i.e., if the center of G is compact). In this case, its sign is $(-1)^\ell$, where ℓ is the rank of L over k .*

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Let $G' \rightarrow G$ be an *isogeny* and let n (resp. c) be the number of elements of its kernel (resp. the index of its image). As G and G' are locally isomorphic, the canonical measure μ_G of G defines a measure $(\mu_G)'$. An easy calculation shows that *the canonical measure $\mu_{G'}$ of G' is equal to*

$$\frac{c}{n}(\mu_G)';$$

it is therefore zero (resp. positive, negative) if and only if μ_G is.

On the other hand, if $L = L_1 \times L_2$ and if $G = G_1 \times G_2$ is the corresponding decomposition of G , we have $\mu_G = \mu_{G_1} \otimes \mu_{G_2}$, which is immediate.

Combining the two facts above, we are reduced to verifying the proposition in the following three special cases:

- a) L is simple and simply connected;
- b) $L = \mathbf{G}_m$;
- c) L is a torus of rank 0 over k .

The first case has been treated above; in the second, we have $\mu_G = 0$ and in the third G is compact and μ_G is the canonical Haar measure of G . Whence the proposition.

3.5. Passage to products

Let $(G_\alpha)_{\alpha \in A}$ be a finite family of Lie groups. We assume that each G_α is of one of the following types (cf. no. 2.3):

- (i) a real unimodular Lie group having a finite number of connected components;
- (ii) the group $L_\alpha(k_\alpha)$ of k_α -points of a connected reductive algebraic group L_α over a locally compact local field k_α .

We denote by G_∞ (resp. G_f) the product of the G_α of type (i) (resp. of type (ii)), and we set

$$G = \prod_{\alpha \in A} G_\alpha = G_\infty \times G_f.$$

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The groups G_∞ , G_f and G are unimodular locally compact groups. It is known (Prop. 22) that G_∞ possesses an Euler-Poincaré measure; let μ_∞ be such a measure. On the other hand, if G_α is of type (ii),

we defined in no. 3.3 a canonical measure μ_{G_α} on G_α ; by passing to products, this defines a measure μ_f on G_f . Hence finally a measure

$$\mu = \mu_\infty \otimes \mu_f$$

on $G = G_\infty \times G_f$.

Theorem (7). *The measure μ defined above is an Euler-Poincaré measure on G .*

We argue by induction on the number of G_α of type (ii) and apply the corollary to Prop. 25 in no. 3.3.

Corollary. *Let Γ and Γ' be two torsion-free discrete subgroups of G , with compact quotient. If $\mu \neq 0$, the ratio of the volumes of G/Γ and G/Γ' is a rational number.*

Indeed, this ratio is equal to $\mu(G/\Gamma)/\mu(G/\Gamma') = \chi(\Gamma)/\chi(\Gamma')$.

Remarks

- 1) When G_∞ is reductive, Prop. 23 and 28 make it possible to determine if the measure μ is > 0 , zero, or < 0 .
- 2) If $\mu = 0$, almost nothing is known about the arithmetic nature of the volumes of compact G/Γ . For example, let Γ and Γ' be two discrete subgroups with compact quotient of $G = \mathrm{SL}_n(\mathbf{R})$, $n \geq 3$. Is it true that the ratio of the volumes of G/Γ and G/Γ' is a rational number? This is very unlikely, but no counterexample is known; the question is related to that of the values of zeta functions at odd positive integers.

Question

Let Γ be a torsion-free discrete subgroup of G such that G/Γ is

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of finite volume, but not necessarily compact. Is it true that Γ is of type (FL) and that $\chi(\Gamma) = \mu(G/\Gamma)$? This seems probable, given the recent results of Harder on arithmetic groups (cf. no. 3.6). It would suffice in any case to solve the question when G is a real Lie group, Prop. 25 allowing one to pass from there to the general case.

3.6. S -arithmetic subgroups.

Let us recall the notation and hypotheses of no. 2.4: L is a reductive algebraic group over a global field k , and S is a finite non-empty set of places of k containing the archimedean places. For every $v \in S$, we denote by k_v the completion of k at v , and we set

$$G = \prod_{v \in S} G_v \quad \text{where} \quad G_v = L(k_v).$$

Let μ be the measure defined in the preceding no. on the group G .

Theorem (8). *Let Γ be an S -arithmetic subgroup of $L(k)$. If the rank of L over k is zero, we have $\chi(\Gamma) = \mu(G/\Gamma)$.*

(The expression $\chi(\Gamma)$ makes sense because Γ is of type (WFL) according to Thm. 4 of no. 2.4.) This follows from Thm. 7, since we know that G/Γ is compact.

In the corollaries below, we assume L is semisimple.

Corollary (1). *If all the G_v corresponding to the archimedean places are compact, we have $\chi(\Gamma) \neq 0$, and the sign of $\chi(\Gamma)$ is $(-1)^{d(G)}$.*

(This applies in particular when k is a function field over a finite field.)

The archimedean factor G_∞ of G is compact; its Euler-Poincaré measure is therefore > 0 (and indeed of mass 1). On the other hand, if v is ultrametric, Prop. 28 shows that the canonical measure of G_v is non-zero, and that its sign is $(-1)^{d(G_v)}$, where $d(G_v)$ denotes the rank of L over k_v ; the Euler-Poincaré measure of G is therefore $\neq 0$, and its sign is $(-1)^{d(G)}$,

$$(-1)^{d(G)}, \quad \text{with} \quad d(G) = \sum_{v \in S} d(G_v). \quad \text{Hence the corollary.}$$

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COROLLARY 2. *If one of the places of k is complex, and if $\dim(L) \geq 1$, we have $\chi(\Gamma) = 0$.*

Indeed, we have seen (cf. no. 3.2, example 2) that the Euler-Poincaré measure of $L(\mathbb{C})$ is zero; thus, the same holds for that of G .

Semi-simple groups of rank ≥ 1

Suppose now that L is semi-simple and of rank ≥ 1 over k . Let Γ be an S -arithmetic subgroup of $L(k)$. The quotient G/Γ is not compact, but it has finite volume. If μ denotes, as above, the canonical measure of G , the real number $\mu(G/\Gamma)$ has a meaning. It is necessary to interpret it. Let us distinguish two cases:

i) *k is a function field over a finite field*

The existence in L of subgroups isomorphic to the additive group $\mathbf{G}_a$ shows that Γ is not of type (WFL) and the definitions of §1 no longer apply; one cannot speak of $\chi(\Gamma)$. To recover finiteness properties, it is probably necessary to introduce a relative cohomology measuring in some sense the deviation between Γ and its unipotent subgroups (for example); one can then hope that, if Γ is p' -torsion-free, $\chi(\Gamma)$ is equal to the Euler-Poincaré characteristic of this relative cohomology (which would show in particular that $\chi(\Gamma)$ is an integer); in any case, this is how things work for SL_2 , cf. [38], chap. II, §2.

ii) *k is an algebraic number field*

Here the situation is satisfactory: the group Γ is of type (WFL) and we have

$$\chi(\Gamma) = \mu(G/\Gamma)$$

according to a recent result⁸ of Harder [19]. [When $S = \Sigma^\infty$ and Γ

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is torsion-free, Harder shows that the manifold $X_\Gamma = \Gamma \backslash T$ is an increasing union of compact pieces $A(t)$ ($t \rightarrow \infty$), having the same homotopy type as X_Γ , and possessing the following property: if we write the Gauss-Bonnet formula⁹ for $A(t)$

$$\chi(A(t)) = \int_{A(t)} \Omega + \int_{\partial A(t)} \Pi, \quad \text{cf. no. 3.2,}$$

the term coming from the boundary $\int_{\partial A(t)} \Pi$ tends to zero as $t \rightarrow \infty$. By passing to the limit, he deduces that

$$\chi(\Gamma) = \chi(X_\Gamma) = \int_{X_\Gamma} \Omega,$$

hence the desired formula $\chi(\Gamma) = \mu(G/\Gamma)$ in the case under consideration. [The general case is reduced to this one by means of Prop. 25.]

When L is split, Harder also gives an explicit formula for $\chi(\Gamma)$, see below.

⁸Various particular cases of this result were known, notably the one where L is an orthogonal group (Siegel [40], t. III, p. 453–456).

3.7. Euler-Poincaré characteristics and values of zeta functions

We retain the notation and hypotheses of the preceding section, and assume that k is a totally real number field (i.e., all k_v , $v \in \Sigma^\infty$, are isomorphic to $\mathbf{R}$). Let r be the degree of k over $\mathbf{Q}$, and d the absolute value of its discriminant. Let S be a finite subset of Σ containing Σ^∞ and let O_S be the ring of elements of k which are integers at all places of $\Sigma - S$. We define the zeta function of k relative to S by the Euler product

$$\zeta_{k,S}(s) = \prod_{v \in \Sigma - S} \frac{1}{1 - Nv^{-s}} ,$$

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where Nv denotes the number of elements of the residue field of k_v . The function $\zeta_{k,S}$ differs from the usual zeta function ζ_k of the field k only by the omission of the local factors relative to $S - \Sigma^\infty$. It extends to a meromorphic function on the entire complex plane. According to a theorem of Siegel ([40], vol. I, p. 545-546 — see also Klingen [24]) the numbers

$$d^{-1/2} \pi^{-rn} \zeta_{k,S}(n), \quad n \text{ even integer } \geq 2,$$

are *rational*; in view of the functional equation of ζ_k , this is equivalent to saying that $\zeta_{k,S}(s)$ takes *rational values* when s is an integer < 0 , values which are $\neq 0$ if and only if the integer s is *odd*. As Siegel has shown on various examples, these rational values can be interpreted in terms of *volumes* of arithmetic groups, and therefore also of *Euler-Poincaré characteristics*. Let us indicate this interpretation, restricting ourselves to the split case, which has just been completely treated by Harder:

Let L_0 be a simple, simply connected, and split group scheme over the ring O_S , let $L = L_0 \times_{O_S} k$ and let $\Gamma_S = L_0(O_S)$ be the group of O_S -integral points of L_0 ; the group Γ_S is an S -arithmetic subgroup of $L(k)$.

On the other hand, let ℓ , R , W_R be the rank of L , its root system, and its Weyl group; let $m_1, \dots, m_\ell$ be the exponents of W_R ([9], p. 118); let W_K be the Weyl group of a maximal compact subgroup K of $L(\mathbf{R})$ and set

$$c = \text{Card}(W_R)/2^\ell \text{Card}(W_K).$$

We then have (Harder [19], no. 2.2):

$$\chi(\Gamma_S) = \mu(G/\Gamma_S) = c^r \prod_{i=1}^{i=\ell} \zeta_{k,S}(-m_i). \quad (*)$$

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[According to Langlands [26], the *Tamagawa number* of $\mathbf{L}$ is equal to 1. One easily deduces that $\omega(G/\Gamma_S)$ is equal to the product of the $\zeta_{k,S}(1 + m_i)$, ω being a certain “arithmetic” measure on G . It is then necessary to determine the ratio μ/ω ; this is what Harder does, *loc. cit.*, when $S = \Sigma^\infty$; the general case reduces to this one by using Thm. 7 of no. 3.4. One thus obtains an expression for $\mu(G/\Gamma_S)$ in terms of the $\zeta_{k,S}(1 + m_i)$; by applying the functional equation of $\zeta_{k,S}$, one deduces⁴ the formula (*).]

Examples

(i) If $\mathbf{L}$ is of type A_ℓ ($\ell \geq 2$), D_ℓ (ℓ odd ≥ 3) or E_6 , the group $W_{\mathbf{R}}$ does not contain -1 , and the canonical measure μ is zero, cf. no. 3.2; on the other hand, one of the exponents m_i is even ([9], p. 123) so that the corresponding number $\zeta_{k,S}(-m_i)$ is zero. One thus finds (in two different ways) the formula $\chi(\Gamma_S) = 0$.

⁹We assume here that the symmetric space T is of even dimension. When it is of odd dimension, the reasoning is analogous: we replace Ω by 0 and Π by half of the Gauss-Bonnet form of $\partial A(t)$.

⁴When k is a *function field* over a finite field, the same method shows that $\mu(G/\Gamma_S) = \tau \prod_{i=1}^{i=\ell} \zeta_{k,S}(-m_i)$, where τ is the Tamagawa number of $\mathbf{L}$.

(ii) If $\mathbf{L}$ is of type F_4 , the maximal compact subgroup K of $\mathbf{L}(\mathbf{R})$ is of type D_4 ; we have $\text{Card}(W_{\mathbf{R}})/\text{Card}(W_K) = 6$, $c = 3/8$, and $(*)$ is written:

$$\chi(\Gamma_S) = (3/8)^r \zeta_{k,S}(-1) \zeta_{k,S}(-5) \zeta_{k,S}(-7) \zeta_{k,S}(-11).$$

(iii) If $\mathbf{L}$ is of type C_ℓ , we have $\mathbf{L}(\mathbf{R}) = \text{Sp}_{2\ell}(\mathbf{R})$, $K = \text{U}_\ell(\mathbf{C})$, $c = 1$, and $\{m_1, \dots, m_\ell\} = \{1, 3, \dots, 2\ell-1\}$. Whence

$$\chi(\text{Sp}_{2\ell}(O_S)) = \zeta_{k,S}(-1) \zeta_{k,S}(-3) \dots \zeta_{k,S}(-2\ell+1).$$

For example:

$$\chi(\text{Sp}_4(\mathbf{Z}[\frac{1}{p}])) = (1-p)(1-p^3) \zeta(-1) \zeta(-3) = -\frac{(p-1)(p^3-1)}{2^5 3^2 5}.$$

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(iv) Applying (iii) to the case $\ell = 1$, we obtain the following result, which generalizes that of no. 1.8, relating to $k = \mathbf{Q}$:

$$\chi(\text{SL}_2(O_S)) = \zeta_{k,S}(-1) = \zeta_k(-1) \prod_{v \in S - \Sigma^\infty} (1 - Nv). \quad (**)$$

[Note that, in this case, the determination of the canonical measure μ offers no difficulties (cf. nos. 3.1 and 3.2), nor does that of the Tamagawa number ([50], no. 3.4); the only slightly delicate point is the equality $\chi(\Gamma_S) = \mu(G/\Gamma_S)$, which can be verified by means of the compactification given in [7], §17.]

Estimation of the denominator of $\zeta_k(-1)$

The formula $(**)$ above makes it possible to bound the denominator of the rational number $\zeta_{k,S}(-1)$; indeed, if Γ is a torsion-free subgroup of $\text{SL}_2(O_S)$ of finite index N , we have

$$N \cdot \zeta_{k,S}(-1) = N \cdot \chi(\text{SL}_2(O_S)) = \chi(\Gamma),$$

and $\chi(\Gamma)$ is an integer since Γ is of type (FL), cf. Thm. 4. Thus, the denominator of $\zeta_{k,S}(-1)$ is a *divisor* of N . By choosing Γ suitably, one can obtain from this a fairly precise estimate of the denominator in question:

Let ℓ be a prime number. If n is an integer ≥ 0 , let F_n be the field of ℓ^n -th roots of unity, and let E_n be the maximal real subfield of F_n . Let us identify k with a subfield of $\mathbf{R}$, and let $n(\ell)$ denote the largest value of n such that E_n is contained in k (or that $[F_n k : k] \leq 2$, which amounts to the same thing). We have $n(\ell) = 0$ for almost all ℓ . Let us set

$$w = 2^{n(2)+1} \prod_{\ell \neq 2} \ell^{n(\ell)}.$$

PROPOSITION 29. *The product $w \cdot \zeta_{k,S}(-1)$ is an integer.*

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Let ℓ be a prime number, and let $n = n(\ell)$; set $m = n$ if $\ell \neq 2$ and $m = n + 1$ if $\ell = 2$. In view of the argument given above, we are reduced to proving the following result:

Lemma (8). *There exists a torsion-free subgroup of $\text{SL}_2(O_S)$ whose index is finite and is not divisible by ℓ^{m+1} .*

(The “torsion-free” hypothesis could be weakened to “without ℓ -torsion”, cf. no. 1.8, prop. 13.)

Let i be an integer ≥ 0 , and let $\text{Gal}(F_i/\mathbf{Q})$ be the Galois group of $F_i/\mathbf{Q}$. It is known that this group is identified with $(\mathbf{Z}/\ell^i \mathbf{Z})^*$, and that this identification transforms $\text{Gal}(F_i/E_i)$ into $\{\pm 1\}$. Let V_i be the subgroup of $(\mathbf{Z}/\ell^i \mathbf{Z})^*$ corresponding to $F_i \cap k$. We have

$$V_i = \text{Gal}(F_i/(F_i \cap k)) = \text{Gal}(F_i k/k).$$

Since k is real, -1 belongs to V_i . In view of the definition of n , we have $V_i = \{\pm 1\}$ if and only if $i \leq n$. There exists therefore an element $s \in V_{n+1}$ who is distinct from ± 1 . Let Σ_ℓ be the set of $v \in \Sigma - \Sigma^\infty$ whose residue characteristic p_v is distinct from ℓ . If $v \in \Sigma_\ell$, v is unramified in $F_i k$ and the corresponding Frobenius substitution $\sigma_v \in V_i$ is the image of Nv in $(\mathbb{Z}/\ell^i \mathbb{Z})^*$. Let us take $i = n + 1$ and apply Čebotarev's density theorem to the extension $F_i k/k$. We deduce that there exist infinitely many $v \in \Sigma_\ell$ such that $\sigma_v = s$. In view of the choice of s , we therefore have

$$Nv \not\equiv \pm 1 \pmod{\ell^{n+1}}.$$

One easily verifies that this implies

$$\begin{aligned} Nv^2 &\not\equiv 1 \pmod{\ell^{n+1}} & \text{if } \ell \neq 2 \\ Nv^2 &\not\equiv 1 \pmod{2^{n+2}} & \text{if } \ell = 2. \end{aligned} \tag{***}$$

Let us choose such a v , and suppose furthermore that $v \notin S$, that v is non-

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ramified over $\mathbf{Q}$, and that $p_v \neq 2$; this is possible since it only excludes a finite number of v . Let $\mathfrak{p}_v$ be the maximal ideal of O_S defined by v , and let Γ_v be the subgroup of $\mathrm{SL}_2(O_S)$ formed by the matrices congruent to 1 modulo $\mathfrak{p}_v$. A classical argument, due to Minkowski [31], shows that the congruence group Γ_v is *torsion-free*. Its index is $Nv(Nv^2 - 1)$; in view of (***) above, it is not divisible by ℓ^{m+1} , which proves the lemma.

REMARK

According to a conjecture of Bass, Birch, and Tate, the absolute value of the integer $w\zeta_k(-1)$ of Prop. 23 should be equal to the order of a certain subgroup of the “ K_2 ” of the field k (the intersection of the kernels of the “tame symbols”). As Tate remarked, this conjecture suggests various divisibility properties of $w\zeta_k(-1)$, which one can try to interpret in terms of Euler-Poincaré characteristics. Here is an example:

PROPOSITION 30. *Let S be the union of Σ^∞ and the ultrametric places v such that $p_v = 2$. Let C_S be the group of ideal classes of O_S of square equal to 1. Set*

$$\mathrm{Card}(O_S^*/O_S^{*2}) = 2^e \quad \text{and} \quad \mathrm{Card}(C_S) = 2^c.$$

The integer $w\zeta_k(-1)$ is divisible by 2^{e+c-1} .

Note that, by the unit theorem, we have $e = \mathrm{Card}(S) \geq r + 1$. Whence the following result, also proved by Harder and Hirzebruch:

COROLLARY. *We have $w\zeta_k(-1) \equiv 0 \pmod{2^r}$, where $r = [k : \mathbf{Q}]$.*

The proof of Proposition 30 uses a certain S -arithmetic subgroup of the adjoint group $\mathrm{PGL}_2 = \mathrm{SL}_2/\{\pm 1\}$. More precisely, let Λ be the lattice $O_S \oplus O_S$ of k^2 , so that $\mathrm{SL}(\Lambda) = \mathrm{SL}_2(O_S) = \Gamma$; let Γ' be the subgroup $\Gamma/\{\pm 1\}$ of $\mathrm{PGL}_2(k)$. On the other hand, let $\tilde{\Gamma}' \dots$

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the subgroup of $\mathrm{GL}_2(k)$ consisting of elements g such that there exists an ideal $\mathfrak{a}$ of O_S with $g(\Lambda) = \mathfrak{a} \cdot \Lambda$; the group $\tilde{\Gamma}''$ contains k^* ; let $\Gamma'' = \tilde{\Gamma}''/k^*$ be its image in $\mathrm{PGL}_2(k)$. We have $\Gamma' \subset \Gamma''$; moreover, Γ'' is identified with the group of automorphisms of the matrix algebra $M_2(O_S)$, hence is an S -arithmetic subgroup of $\mathrm{PGL}_2(k)$. Reasoning as in the proof of Prop. 29, we deduce that the product $w\chi(\Gamma'')$ is an integer (note that, for any finite field $\bar{k}$, $\mathrm{SL}_2(\bar{k})$ and $\mathrm{PGL}_2(\bar{k})$ have the same number of elements).

On the other hand, let $g \in \tilde{\Gamma}''$, and let $c(g)$ be the class of the ideal $\mathfrak{a}$ such that $g(\Lambda) = \mathfrak{a} \cdot \Lambda$; passing to the quotient, the map $g \mapsto c(g)$ defines a homomorphism from Γ''/Γ' onto the group C_S , and the kernel of this homomorphism is isomorphic to O_S^*/O_S^{*2} . We conclude that $(\Gamma'' : \Gamma') = 2^{e+c}$, whence $\chi(\Gamma'') = \chi(\Gamma')/2^{e+c}$. But any torsion-free subgroup of finite index N in Γ is isomorphic to its image in Γ' , which is of index $N/2$ in Γ' . We deduce that $\chi(\Gamma') = 2\chi(\Gamma) = 2\tilde{\chi}_{k,S}(-1)$, whence

$$\chi(\Gamma'') = \tilde{\chi}_{k,S}(-1)/2^{e+c-1}.$$

Since $w\chi(\Gamma'')$ is an integer, this shows that $w\tilde{\zeta}_{k,S}(-1)$ is divisible by 2^{e+c-1} . But $\tilde{\zeta}_{k,S}(-1)$ differs from $\tilde{\zeta}_k(-1)$ by the factor $\prod_{p_v|2}(1-Nv)$, which is an odd integer. We thus see that $w\tilde{\zeta}_k(-1)$ is divisible by 2^{e+c-1} , whence the proposition.

Examples

- i) For $k = \mathbf{Q}$, we have $w = 2^3 \cdot 3$, $\zeta(-1) = -1/12$, whence $w\zeta(-1) = -2$, which is indeed divisible by $2^{e+c-1} = 2$.
- ii) For $k = \mathbf{Q}(\sqrt{5})$, we have $w = 2^3 \cdot 3 \cdot 5$ and $\zeta_k(2) = 2\pi^4/75\sqrt{5}$, cf. Siegel [41], p. 32. The formula

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$$\zeta_k(-1) = d^{3/2}\zeta_k(2)/(-2\pi^2)^r$$

then shows that $\zeta_k(-1) = 1/(2 \cdot 2 \cdot 5)$ whence $w\zeta_k(-1) = 2^2$, which is indeed divisible by $2^{e+c-1} = 2^2$.

- iii) For $k = \mathbf{Q}(\sqrt{15})$, we likewise find $w\zeta_k(-1) = 2^4 \cdot 3$, which is indeed divisible by $2^{e+c-1} = 2^2$.

iv) Let us take for k the maximal real subfield of the field of 7-th roots of unity; this is a cyclic cubic field. We have $w = 2^3 \cdot 3 \cdot 7$ and $\zeta_k(2) = 2^3\pi^6/(3 \cdot 7^4)$, cf. [41], p. 100-101. From this we obtain $\zeta_k(-1) = -1/(3 \cdot 7)$, whence $w\zeta_k(-1) = -2^3$ which is indeed divisible by $2^{e+c-1} = 2^3$.

Denominators of $\zeta_k(1-2n)$

One can apply to the group Sp_{2n} the same method as above. The index of the congruence subgroup associated with an ultrametric place v is equal to $Nv^{n^2} \prod_{i=1}^{i=n} (Nv^{2i} - 1)$. One deduces from this that the product of the $(Nv^{2i} - 1)\zeta_k(1-2i)$, $1 \leq i \leq n$, has as denominator a power of p_v . For any prime number ℓ , and any even integer $2i$, let us then denote by $m(i, \ell)$ the largest integer m such that ℓ^m divides all the $(Nv^{2i} - 1)$, with $p_v \neq \ell$, and let w_i be the product of the $\ell^{m(i, \ell)}$. Reasoning as in the proof of Lemma 8, we obtain:

Proposition (31). *For any $n \geq 1$, the rational number $\prod_{i=1}^{i=n} w_i \zeta_k(1-2i)$ is an integer.*

It is easy to explicitly determine the w_i as a function of the intersection of k with the field of all roots of unity; we leave this task to the reader.

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Questions

1) Is it true that each of the $w_i \zeta_k(1-2i)$ is an integer? In other words, if v is an ultrametric place of k , of residue characteristic p , is it true that the product of $\zeta_k(1-2i)$ by $(Nv^{2i} - 1)$ belongs to $\mathbf{Z}[\frac{1}{p}]$? If so, can one interpret the integers $w_i \zeta_k(1-2i)$ in terms of Euler-Poincaré characteristics?

2) Are there congruences relating the $w_i \zeta_k(1-2i)$ for different values of i , and generalizing Kummer's congruences on Bernoulli numbers? The question is related to the very mysterious one of a possible extension of the results of Kubota-Leopoldt to all totally real fields.

3) Can one interpret in terms of Euler-Poincaré characteristics the values at negative integers of the zeta functions relative to the *ideal classes* (modulo a conductor) of any number field? We know that these values are rational and we have an upper bound on their denominators (Siegel [41]); we can hope to improve it.

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